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Feature Learning for matching between microscopy data

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Feature Learning for matching between microscopy data:

The precise matching of individual cells in 3D microscopy images is a fundamental challenge in developmental biology and systems neuroscience. The nematode *C. elegans*, with its invariant cell lineage, serves as an ideal model system for this task. While traditional methods achieve high accuracy, they predominantly rely on hand-crafted features, which limits their adaptability and potential for end-to-end optimization.

This thesis introduces a novel, end-to-end deep learning framework for automated cell matching that addresses these limitations. The framework integrates a 3D Convolutional Autoencoder for unsupervised feature extraction from raw voxel data with a Graph Attention Network (GAT) to model the global spatial relationships between cells. This hybrid approach effectively fuses high-dimensional, learned morphological features with low-dimensional, explicit geometric features to create a discriminative representation for cell correspondence.

Our model achieves a matching accuracy of 74.09% on a *C. elegans* dataset. Extensive experiments demonstrate that integrating the GAT yields a significant performance increase of over 56 percentage points and that L1 loss consistently outperforms MSE loss for this matching task. Although a performance gap to state-of-the-art methods remains, this work establishes a robust baseline for pure deep learning approaches in cellular correspondence. It provides systematic design principles for model architecture and loss function selection, paving the way for future advancements in automated biological image analysis.

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1 Introduction

The accurate identification and matching of individual cells across different biological specimens is a fundamental challenge in developmental biology and systems neuroscience. *Caenorhabditis elegans* (*C. elegans*) presents an exceptional model for this task; with its invariant cell lineage of exactly 558 somatic cells in the L1 larval stage and a completely mapped connectome, it provides the ground truth necessary to develop and validate scalable, automated solutions for cellular correspondence.

Despite the biological advantages of *C. elegans*, achieving robust cell matching in 3D microscopy images remains computationally demanding. While state-of-the-art methods like RAPCAT (95.40% accuracy) and Active Graph Matching (92.00%) have demonstrated remarkable success, they predominantly rely on hand-crafted geometric features. This paradigm, while effective, faces inherent limitations in capturing complex cellular morphologies and generalizing across diverse imaging conditions without significant manual re-engineering. This reliance on predefined features creates a critical gap for more adaptive, data-driven approaches.

To address this gap, this thesis introduces a novel, end-to-end deep learning framework that integrates 3D Convolutional Neural Networks (CNNs) with Graph Attention Networks (GATs) for automated cell matching. Our approach learns discriminative feature representations directly from raw voxel data, combining local cellular morphology learned from 32^3 voxel subvolumes with global spatial context modeled by the graph network. This allows for joint optimization of the feature extraction and matching processes. Furthermore, we initiate an exploration into rotation-invariant features using group equivariant convolutions to tackle variations in sample orientation.

Our work achieves a matching accuracy of 74.09% on the *C. elegans* dataset, establishing a robust baseline for end-to-end deep learning in this domain. The primary contributions of this research are threefold:

1. **An Integrated End-to-End Architecture:** We design and validate a pipeline that jointly learns and optimizes feature extraction and cell matching, demonstrating that a 3D CNN combined with a GAT can improve accuracy by over 56 percentage points by capturing critical inter-cellular relationships.
2. **A Multi-modal Feature Fusion Strategy:** We effectively fuse high-dimensional learned features with low-dimensional geometric features, creating a more discriminative and biologically meaningful representation for cell correspondence.
3. **Systematic Design Principles for Cell Matching Models:** Through extensive ablation studies, we provide clear architectural guidelines, empirically

demonstrating the superiority of Residual connections and L1 loss for this task, offering valuable insights for future model development.

The remainder of this thesis is organized as follows: Chapter 2 provides the technical background on graph-based methods and 3D CNNs. Chapter 3 details our proposed methodology. Chapter 4 presents our comprehensive experimental evaluation. Finally, Chapter 5 concludes with a discussion of the research implications and a roadmap for future work.

2 Background

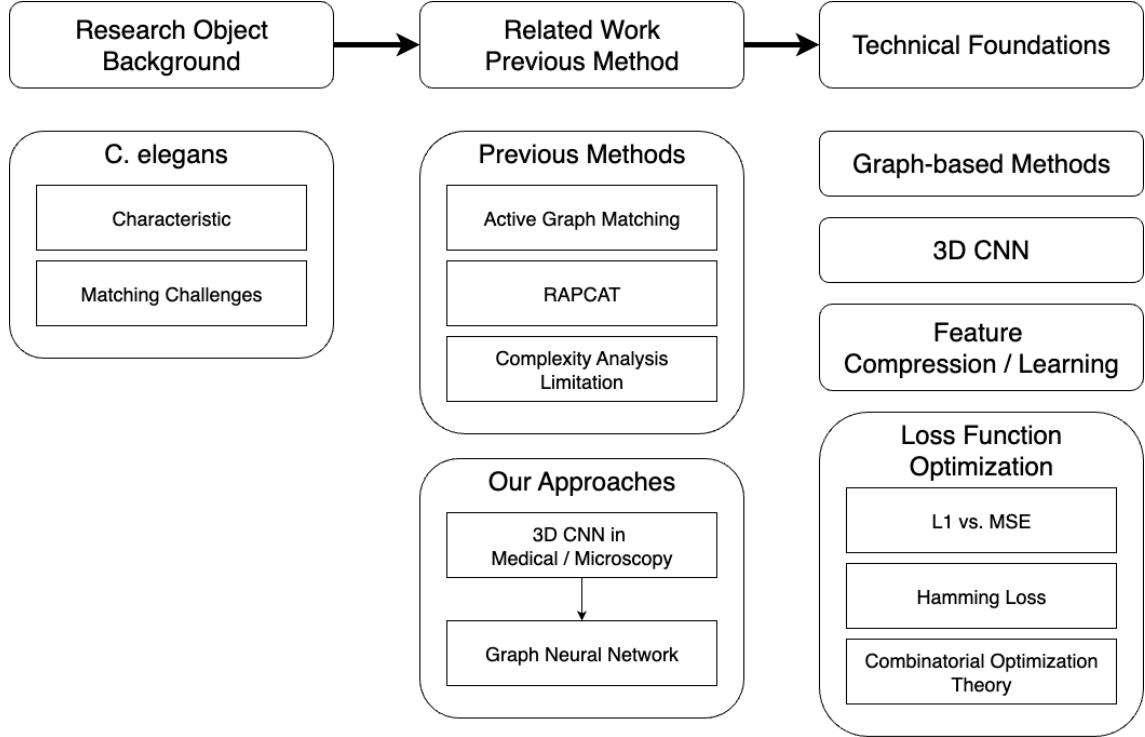


Figure 2.1: An overview of the chapter’s structure, which is divided into three parts: (1) introduction to the model organism *C. elegans* and the challenges of cell matching in its imagery; (2) a review of previous methodologies and the presentation of our proposed approach; and (3) an outline of the key technical foundations that support our work.

2.1 Research Background and Motivation

2.1.1 *C. elegans* as a Model Organism

Caenorhabditis elegans represents an exceptional model organism for automated cell matching and annotation research, offering unique biological characteristics that directly address fundamental challenges in computational biology. Since its introduction as a model organism for neural development studies by Sydney Brenner in 1963, this free-living, transparent nematode has become an indispensable tool in neuroscience, genomics, cell biology, and aging research. Its combination of biological

simplicity and experimental tractability provides an ideal platform for developing and validating computational approaches to cellular analysis.

Invariant Cell Lineage: A Foundation for Automated Cell Matching

The most remarkable feature of *C. elegans* for computational cell matching research is its invariant cell lineage. From fertilized egg to adult, the division, migration, and ultimate fate of every cell follow a nearly identical pattern across individuals. An adult hermaphrodite contains approximately 1,000 somatic cells, with the deterministic nature of development providing unprecedented precision for studying cell differentiation, apoptosis, and tissue formation. This stereotyped development pattern establishes the ground truth necessary for training and validating automated cell matching algorithms, as each cell's identity and spatial position can be precisely determined across different organisms.

The nervous system exhibits similar stereotypy, comprising exactly 302 neurons in the hermaphrodite [Altun et al. \[2001-2024b\]](#). This structural consistency enables researchers to analyze neural circuits at single-cell resolution with remarkable precision. Each *C. elegans* neuron follows a systematic naming convention consisting of one to three uppercase letters indicating anatomical class and synaptic connectivity, often followed by directional suffixes (L/R/D/V for left/right/dorsal/ventral) [Altun et al. \[2001-2024a\]](#). This standardized nomenclature extends to non-neuronal cells, providing a comprehensive framework for cellular identification:

RIMR (Ring Motor neuron, Right): Key motor neurons in the nerve ring involved in sensory integration and head movement modulation, representing critical neural network hubs [Altun et al. \[2001-2024a\]](#)

EXC (Excretory Canal cell): Specialized non-neuronal cells forming the excretory system, illustrating the systematic naming across all functional cell types [Altun et al. \[2001-2024a\]](#)

Complete Connectome and Systematic Cell Mapping

C. elegans achieved the distinction of being the first multicellular organism with a completely mapped neural connectome. This pioneering work identified approximately 5,000 chemical synapses and 600 gap junctions among its 302 neurons, providing the scientific community with the first complete neural circuit diagram [Altun et al. \[2001-2024b\]](#). This comprehensive mapping establishes a gold standard for validating computational cell identification methods, as every cellular connection and spatial relationship has been experimentally verified.

Experimental Advantages for Computational Research

Several characteristics make *C. elegans* particularly suitable for developing automated cell matching algorithms:

- **Optical Transparency:** The transparent body enables direct observation of biological processes in vivo using standard light microscopy. Combined with fluorescent tagging techniques like Green Fluorescent Protein (GFP), researchers can track protein expression, cell division, and neuronal activity in real-time without invasive procedures.
- **Genetic Tractability:** As the first multicellular organism with a completely sequenced genome, *C. elegans* offers extensive genetic tools. A significant portion of its genes has functional homologs in humans, while techniques such as RNA interference (RNAi) enable efficient genetic manipulation for functional studies.
- **Experimental Efficiency:** The extremely short life cycle (egg to adult in approximately three days) and ease of large-scale laboratory cultivation make *C. elegans* ideal for high-throughput computational studies. Strains can be cryopreserved indefinitely, ensuring experimental reproducibility across studies.

Dataset Characteristics and Research Applications

This research utilizes a comprehensive dataset comprising high-resolution 3D confocal fluorescence microscopy images of *C. elegans* larvae at the L1 developmental stage [Taylor et al. \[2024a,b\]](#). Each larva contains exactly 558 somatic cells [Long et al. \[2009\]](#), all annotated using the machine learning-based RAPCAT (Registration and Automated Prediction of Cell Annotations Tool) [Taylor et al. \[2024b\]](#). The annotation pipeline integrates nuclear DAPI staining with histone-tagged fluorescent reporters, enabling precise identification of individual cell identities while preserving 3D structural integrity [Taylor et al. \[2024a,b\]](#).

The dataset encompasses over 620 transgenic *C. elegans* strains, each engineered to express distinct transcription factors fused to fluorescent proteins (GFP or mCherry). This multiplexed reporter system enables quantitative profiling of transcription factor expression at single-cell resolution, supporting comprehensive spatially resolved gene expression mapping and system-wide cellular annotation at unprecedented scale [Taylor et al. \[2024a,b\]](#).

The deterministic cellular development, complete connectome mapping, and extensive genetic tools of *C. elegans* provide an ideal biological platform for developing robust computational cell matching methods. However, despite these advantages, automated cell matching in 3D microscopy images presents significant technical challenges that motivate the development of advanced computational approaches.

2.1.2 Cell Matching Challenges in 3D Microscopy

Automated cell matching in 3D microscopy images presents fundamental computational challenges that necessitate sophisticated algorithmic solutions. Despite the stereotyped development of *C. elegans*, establishing accurate cell correspondences

across different organisms and imaging conditions remains technically demanding, driving the need for advanced machine learning approaches [Santella et al. \[2014\]](#), [Murray et al. \[2006\]](#).

High-Dimensional Feature Space and Computational Complexity

The analysis of 3D microscopy datasets generates massive amounts of high-dimensional data characterized by complex spatial relationships that pose significant computational challenges. When processing 558 cells simultaneously within three-dimensional space, traditional graph matching algorithms encounter severe scalability limitations, with computational complexity reaching $O(n^4)$ for second-order approaches [Conte et al. \[2004\]](#). This computational burden becomes prohibitive for large-scale cellular datasets, where memory bandwidth limitations and cache efficiency constraints further exacerbate performance bottlenecks. The curse of dimensionality manifests particularly prominently when algorithms attempt to simultaneously capture local cellular morphological features and preserve global spatial context within reasonable computational constraints [Kroeger et al. \[2018\]](#), [Stegmaier et al. \[2016\]](#).

Morphological Variability and Imaging Artifacts

Morphological variability presents a multifaceted challenge that significantly complicates automated cell matching algorithms. Developmental timing differences across specimens result in cells appearing at slightly different stages, introducing systematic variations in morphological parameters such as cell volume, nuclear-to-cytoplasmic ratios, and fluorescent reporter expression levels [Murray et al. \[2006\]](#). These temporal variations are compounded by systematic imaging artifacts arising from differences in sample preparation protocols, mounting procedures, and microscopy acquisition parameters. Such technical variations introduce confounding factors that significantly impact feature-based matching performance, particularly when combined with inherent optical limitations of 3D fluorescence microscopy [Sage et al. \[2019\]](#).

The anisotropic nature of 3D microscopy data presents additional challenges, where axial resolution typically degrades by a factor of 2-3 compared to lateral resolution. This resolution asymmetry, combined with depth-dependent signal attenuation and spatially varying noise characteristics, necessitates specialized preprocessing and feature extraction methodologies that can accommodate these systematic distortions [Weigert et al. \[2018\]](#).

Scale and Orientation Invariance Requirements

Cellular structures exhibit considerable variability in spatial orientation and scale across different samples, demanding robust feature extraction methods capable of maintaining consistent representations under various geometric transformations. The requirement for rotation invariance becomes particularly challenging in three-dimensional

space, where cells may appear in arbitrary orientations without the benefit of extensive data augmentation strategies [Zhou et al. \[2020a\]](#), [Kreshuk et al. \[2011\]](#). Feature extraction algorithms must therefore incorporate inherent invariance properties while preserving critical spatial relationships between neighboring cellular structures.

Furthermore, biological systems exhibit natural variation in cellular positioning and morphology that algorithms must distinguish from meaningful biological differences. This necessitates the development of feature representations that are robust to irrelevant variations while remaining sensitive to biologically significant cellular characteristics [Caicedo et al. \[2019\]](#), [Al-Masni et al. \[2018\]](#).

Limited Training Data and Ground Truth Validation

The scarcity of expertly annotated 3D cellular datasets poses significant obstacles for supervised machine learning approaches. Manual annotation of cellular structures across three-dimensional volumes requires substantial time investment from domain experts and specialized biological knowledge, resulting in limited dataset sizes relative to the complexity of the matching problem [Maška et al. \[2014\]](#). Inter-annotator variability further complicates ground truth establishment, particularly for cells located in densely packed regions or at tissue boundaries where expert judgments may differ substantially.

These limitations increase the risk of model overfitting, especially given the high-dimensional nature of the feature space relative to available training samples. Consequently, successful approaches must incorporate careful regularization strategies and robust cross-validation methodologies to ensure generalization performance [Ronneberger et al. \[2015b\]](#).

2.2 Related Work and Previous Methods

2.2.1 Previous Cell Annotation Methods

The development of automated cell annotation methods has evolved from simple geometric matching to sophisticated graph-based approaches. Early methods relied primarily on morphological features and spatial proximity, but the complexity of 3D cellular structures necessitated more advanced algorithmic frameworks that could capture both local cellular characteristics and global spatial relationships.

Active Graph Matching

Active Graph Matching represents an innovative approach for automatic segmentation and annotation of *C. elegans* cell nuclei that integrates Active Shape Models into sparse graph matching problems [Kainmueller et al. \[2013\]](#). The core innovation of this method lies in combining the advantages of global statistical deformation models with local deformation models, using second-order random fields

to describe local deformations. The method learns global-plus-local deformation models of *C. elegans* from training data, where the global model serves as a "backbone" of plausible nuclei constellations, while the local model penalizes deviations from this backbone.

The method establishes a three-layer statistical model architecture: first, a global point distribution model that uses principal component analysis to establish a variability model of nuclei center points; second, local point distribution models that establish covariance matrices for each nucleus position; and finally, local shape models that describe the variability of each nucleus shape. In terms of optimization strategy, the method employs iterative optimization techniques that can be viewed as a generalized Iterative Closest Points (ICP) method, where "Closest" is replaced by "Best Matching" in terms of global second-order matching energy.

Experimental results demonstrate that in the task of annotating 357 nuclei in *C. elegans* L1 larvae, this method achieved an average annotation accuracy of 92%, significantly outperforming the state-of-the-art method of 86% at the time. However, the method's reliance on hand-crafted statistical models and its computational complexity limitations restrict its scalability to larger datasets.

RAPCAT Method

The RAPCAT (Robust-point-matching- And Piecewise-affine-based Cell Annotation Tool) developed by [Li et al. \[2024\]](#) represents another important technological breakthrough in automated cell annotation. The technical architecture of this tool comprises four main components: digital worm templates built from 100 manually annotated L1 larval image stacks, with each template containing 3D coordinates of 558 cell points; robust point matching using Robust Point Matching (RPM) methods for initial cell identity assignment; piecewise affine transformation that remaps cell positions in a topology-preserving manner through global and piecewise affine transformations; and bipartite matching improvement using bipartite matching algorithms to ensure that remapped cell positions optimally fit the reference template.

RAPCAT achieved an average accuracy of 95.4% in automatic annotation and provides Cell Annotation Confidence (CAC) scores to predict annotation accuracy. The method represents a significant advancement in terms of both accuracy and computational efficiency compared to earlier approaches. However, RAPCAT still primarily relies on geometric features and hand-crafted transformations, limiting its ability to capture complex cellular morphological patterns that could be learned from data.

Computational Complexity Analysis and Limitations

The computational complexity analysis of existing methods reveals fundamental scalability limitations that motivate the development of more efficient approaches:

Active Graph Matching Complexity The computational complexity of the Active Graph Matching method primarily stems from solving second-order graph matching problems. Given n cells, the method needs to consider all possible cell correspondences, where unary terms have complexity $O(n^2)$ and binary terms have complexity $O(n^4)$. Due to the adoption of second-order random field models to describe local deformations, the graph matching problem in each iteration requires solving using Dual-Decomposition methods, resulting in an overall time complexity of $O(n^4)$, which creates significant computational bottlenecks when processing large-scale cell data.

RAPCAT Complexity The computational complexity of the RAPCAT method is relatively low, divided into three main stages. First, the Robust Point Matching stage has complexity $O(n^2)$, where n is the number of cells. Second, the piecewise affine transformation stage requires transforming local regions with complexity $O(kn)$, where k is the number of segments. Finally, the bipartite matching stage uses the Hungarian algorithm with complexity $O(n^3)$. Therefore, RAPCAT’s overall time complexity is $O(n^3)$, representing a significant improvement over Active Graph Matching.

Linear Assignment Method Comparison Traditional Linear Assignment methods, such as the Hungarian algorithm, have time complexity $O(n^3)$ and space complexity $O(n^2)$. This method only considers one-to-one matching relationships between cells and does not involve high-order spatial constraints, thus achieving high computational efficiency. However, the main limitation of Linear Assignment methods is their reliance solely on cell nucleus center coordinates for matching, lacking consideration of cellular internal structure and texture features, which easily leads to annotation errors in densely packed cell regions (such as brain areas).

In comparison, while Active Graph Matching considers richer spatial relationships, its $O(n^4)$ complexity makes it difficult to scale to large applications. RAPCAT maintains $O(n^3)$ complexity while enhancing spatial constraint expression through piecewise affine transformations, but still primarily relies on geometric features for matching.

Fundamental Limitations These traditional approaches share several fundamental limitations: (1) reliance on hand-crafted features that may not capture optimal cellular representations; (2) inability to perform end-to-end learning that jointly optimizes feature extraction and matching; (3) limited adaptability to different imaging conditions and cellular morphologies; and (4) computational scalability issues for large-scale datasets. These limitations motivate the development of deep learning-based approaches that can learn optimal feature representations while maintaining computational efficiency.

2.2.2 Deep Learning Approaches in Medical Imaging

The application of deep learning to medical imaging has revolutionized automated analysis across numerous domains, providing crucial context for understanding the potential of neural networks in cellular annotation tasks. Recent advances in 3D convolutional neural networks, attention mechanisms, and graph neural networks have demonstrated remarkable success in medical image analysis, establishing the foundation for their application to cellular matching problems.

3D Convolutional Neural Networks in Biomedical Applications 3D CNNs have shown exceptional performance in volumetric medical image analysis, particularly in tasks requiring spatial context understanding. Notable applications include organ segmentation, tumor detection, and cellular structure analysis. The PlantSeg framework [Wolny et al. \[2020a\]](#) demonstrated the effectiveness of 3D U-Net architectures for plant cell segmentation, achieving state-of-the-art results on challenging datasets with anisotropic resolution characteristics similar to those encountered in *C. elegans* imaging.

The success of 3D CNNs in biomedical applications stems from their ability to capture hierarchical spatial features across multiple scales, from local cellular textures to global tissue organization. However, traditional 3D CNNs face challenges in handling rotation invariance and computational efficiency, particularly when processing large volumetric datasets.

Graph Neural Networks in Biological Systems Graph Neural Networks have emerged as powerful tools for modeling biological systems with irregular structures and complex relationships. Applications range from protein structure prediction to cellular interaction modeling, demonstrating the versatility of graph-based approaches in capturing biological complexity. The ability of GNNs to handle variable-sized inputs and incorporate relational information makes them particularly suitable for cellular analysis tasks where spatial relationships are crucial.

Recent work has shown that Graph Attention Networks can effectively model cellular neighborhoods and capture context-dependent relationships that are essential for accurate cell identification. The attention mechanism allows the network to dynamically focus on relevant neighboring cells, adapting to different cellular densities and spatial arrangements.

Integration Challenges and Opportunities Despite the success of deep learning in medical imaging, several challenges remain in applying these methods to cellular matching tasks: (1) the discrete nature of matching problems conflicts with the continuous optimization required by neural networks; (2) limited annotated datasets restrict the training of complex models; (3) the need for biological interpretability constrains architectural choices; and (4) computational efficiency requirements for practical deployment.

These challenges have driven the development of differentiable optimization techniques and hybrid approaches that combine the representational power of deep learning with the structural constraints of graph matching. The integration of these techniques forms the foundation for the methodological contributions of this research.

The limitations of existing traditional methods and the promising developments in deep learning for medical imaging establish a clear motivation for developing novel approaches that combine learned feature representations with principled matching algorithms. This convergence of graph matching theory and deep learning practice provides the foundation for the technical innovations presented in this work.

2.3 Technical Foundations

2.3.1 Graph-based Methods

Graph Matching Theory

Graph matching is a combinatorial optimization problem that seeks to establish node correspondences between two graphs, aiming to minimize the matching cost or maximize structural similarity. Applications include shape correspondence, point cloud registration, activity recognition, and, most notably, **keypoint matching** in images, which is critical for pose estimation and semantic alignment.

Problem Formulation and Mathematical Framework Formally, graph matching can be modeled as either a Linear Assignment Problem (LAP) or a Quadratic Assignment Problem (QAP). LAP considers pairwise node similarity, while QAP incorporates edge structure to preserve graph topology, though it is NP-hard. These models operate over permutation matrices, making the problem inherently discrete and non-differentiable.

This non-differentiability presents a major challenge when integrating graph matching into deep learning frameworks. Since the solution is a permutation matrix—binary and one-to-one—standard backpropagation cannot directly propagate gradients through the matching layer. To address this, [Vlastelica et al. \[2020\]](#) proposed *black-box differentiation of combinatorial solvers*, which allows the use of existing solvers (e.g., Hungarian algorithm) during forward pass and approximates the gradient by perturbing the input cost vector. This method enables embedding non-differentiable solvers into neural networks via a closed-form gradient approximation derived from first-order Taylor expansion assumptions.

Deep Graph Matching Framework Development Building upon this, [Rolínek et al.](#) introduced the *Deep Graph Matching via Blackbox Differentiation* framework, where convolutional neural networks (CNNs) extract local features from images and construct similarity graphs incorporating spatial and topological information. The QAP is then solved using dual decomposition, reducing the original problem into

a series of LAP subproblems. These are iteratively solved to obtain a final permutation matrix that satisfies one-to-one matching. The framework was validated on benchmarks such as SPair-71K and WillowObjectClass, demonstrating strong generalization and practicality.

Later, [Sahoo et al. \[2022\]](#) proposed a simpler differentiable strategy called *Identity with Projection*, which treats the solver as a negative identity mapping during backpropagation and projects the input gradients to stable directions. This removes the need for additional solver queries or hyperparameter tuning, yielding efficient and robust training.

Unsupervised Keypoint Correspondence Applications On the application side, [Tourani et al. \[2024\]](#) presented the method, targeting **unsupervised keypoint correspondence**. Their approach leverages discrete cycle-consistency as a training signal: given matchings P_{12} , P_{23} , and P_{31} between images I_1 , I_2 , and I_3 , it enforces the constraint $P_{31}P_{23}P_{12} \approx I$, where I is the identity matrix. This self-supervised mechanism encourages consistent global matchings and integrates well with both LAP/QAP solvers and neural architectures. Experimental results on Pascal VOC, Willow, and SPair-71K show high matching accuracy under low-supervision regimes.

Differentiation of Blackbox Combinatorial Solvers The core idea of this method is to treat the combinatorial solver as a "blackbox." During the forward pass, any existing, efficient solver is used directly to find the optimal solution $y^*(c)$ for a given cost vector c . Its true innovation lies in the backward pass. The paper demonstrates that for a given loss function L , the gradient with respect to the input cost c , denoted as $\frac{\partial L}{\partial c}$, can be approximated by applying a small perturbation to the cost vector.

This is where the hyperparameter λ (Lambda) comes into play. Specifically, the method computes the gradient by solving a new, perturbed problem. Assuming the gradient from subsequent layers is $g = \frac{\partial L}{\partial y}$, the method calculates a new solution $y_\lambda = \text{solve}(c - \lambda g)$. Consequently, the target gradient $\frac{\partial L}{\partial c}$ can be approximated as:

$$\frac{\partial L}{\partial c} \approx \frac{y^*(c) - y_\lambda}{\lambda}$$

Therefore, λ is a scalar that controls the magnitude of this perturbation and is crucial for the stability and accuracy of the gradient computation. Figure 2.2 visualizes f_λ for different values λ . It is also a key parameter that requires tuning in the subsequent experiments of this thesis.

The advantage of this method lies in its generality: it allows researchers to embed any highly-optimized, off-the-shelf solver for linear objective functions (such as Gurobi or Blossom V) as a module within a neural network. This enables end-to-end learning that integrates feature extraction with combinatorial reasoning, all without needing to modify the solver itself. This technique has not only advanced the field of graph matching but has also opened new avenues for solving other combinatorial optimization problems, such as the Traveling Salesperson Problem and minimum-cost perfect matching.

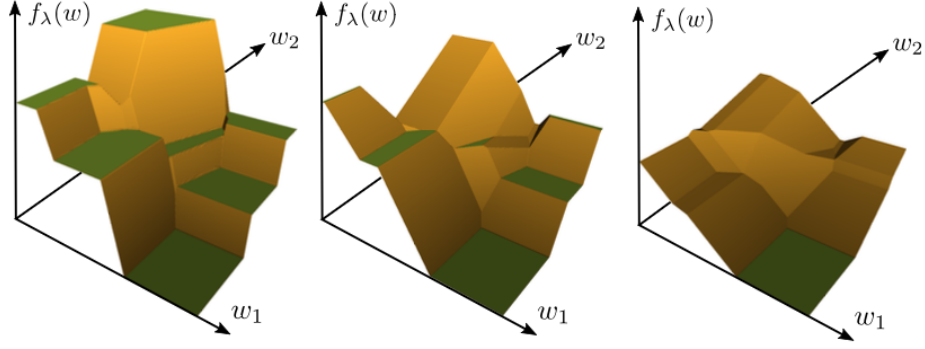


Figure 2.2: Example f_λ for $w \in \mathbb{R}^2$ and $\lambda = 3, 10, 20$ (left to right). As λ changes, the interpolation f_λ is less faithful to the piecewise constant $f(y(w))$ but provides reasonable gradient on a larger set. [Vlastelica et al. \[2020\]](#)

In summary, from its theoretical foundation in black-box differentiability to its practical realization in unsupervised learning frameworks, graph matching has become a crucial component in learning structured correspondences. With further advancements in differentiable solvers and structural loss design, graph matching is poised to play an increasingly central role in cross-domain and weakly supervised learning tasks.

Graph Neural Networks

Graph Neural Networks (GNNs) emerged from the urgent need to extend deep learning to non-Euclidean structured data [Scarselli et al. \[2009\]](#), [Wu et al. \[2020\]](#). Traditional convolutional neural networks excel at processing regular grid data such as images and audio, but vast amounts of real-world data exist in graph structures, including social networks, molecular structures, knowledge graphs, and transportation networks [Zhou et al. \[2020b\]](#). The irregularity and complex inter-node relationships of these data make traditional deep learning methods difficult to apply directly, prompting researchers to explore novel neural network architectures capable of handling graph-structured data [Scarselli et al. \[2009\]](#), [Wu et al. \[2020\]](#). Graph neural networks arose in this context, aiming to extend the success of deep learning to the graph domain and provide powerful tools for processing complex relational data [Zhou et al. \[2020b\]](#).

Graph-structured data possesses a series of unique and challenging characteristics that make the application of traditional machine learning methods complex. First, the number of neighbors for nodes in graphs is variable, unlike images where each pixel has a fixed number of adjacent pixels [Hamilton \[2020\]](#). Second, graph structures lack fixed spatial ordering, making it impossible to define clear sequential relationships as in sequence data processing [Zhou et al. \[2020b\]](#). Furthermore, graph data requires simultaneous processing of joint information from node features and topological structure, demanding models that can effectively integrate local features

and global structural information Scarselli et al. [2009], Hamilton [2020]. These characteristics require new neural network architectures capable of handling irregular data structures while maintaining permutation invariance and locality principles, ensuring model invariance to node reordering and the ability to capture local structural patterns in graphs Zhou et al. [2020b], Hamilton [2020].

Theoretical Foundations Regarding theoretical foundations, a graph $G = (V, E)$ consists of a node set V and an edge set E , with basic mathematical descriptions including the adjacency matrix A and degree matrix D Scarselli et al. [2009]. Elements $A_{ij} = 1$ in the adjacency matrix A indicate connections between nodes i and j , serving as fundamental tools for describing graph topological structure Scarselli et al. [2009]. Spectral graph theory studies the eigenvalues and eigenvectors of the graph Laplacian matrix $L = D - A$, providing frequency domain representations of graphs Hammond et al. [2011], Chung [1997]. In the eigendecomposition $L = U\Lambda U^T$ of the Laplacian matrix, U contains eigenvectors and Λ is the diagonal eigenvalue matrix Hammond et al. [2011], Chung [1997]. Smaller eigenvalues correspond to low-frequency components reflecting global graph structure, while larger eigenvalues correspond to high-frequency components reflecting local variations Hammond et al. [2011], Chung [1997]. This frequency domain understanding provides theoretical foundations for subsequent graph convolution operations Hammond et al. [2011].

Modern GNNs can be understood uniformly under the Message Passing framework, which includes three core steps: message generation $m_{ij} = M(h_i, h_j, e_{ij})$, message aggregation $a_i = \text{Aggregate}(\{m_{ij} : j \in N(i)\})$, and node update $h_i^{(k+1)} = \text{Update}(h_i^{(k)}, a_i)$ Gilmer et al. [2017]. This unified framework provides a clear conceptual foundation for understanding the working principles of different GNN variants Gilmer et al. [2017].

Evolution of Graph Neural Networks The development of graph neural networks can be traced back to the late 1990s, when early research focused on applying recurrent neural networks to directed acyclic graph processing Sperduti and Starita [1997], Sperduti [1997]. In 2009, the pioneering work of Scarselli et al. formally proposed the Graph Neural Network concept, establishing a GNN framework based on fixed-point theory and laying theoretical foundations for subsequent research Scarselli et al. [2009], Gori et al. [2005]. The critical period of modern GNN development began with the breakthrough research of Bruna et al. in 2013, which first successfully extended the concept of convolutional neural networks to graph-structured data, proposing graph convolution methods based on spectral graph theory Bruna et al. [2013].

In 2016, Defferrard et al. proposed ChebNet, innovatively using Chebyshev polynomial approximations of spectral filters, significantly improving computational efficiency and making graph convolution methods applicable to large-scale graphs Defferrard et al. [2016]. The year 2017 marked GNN development entering a new

phase, when Veličković et al. proposed Graph Attention Networks (GAT), introducing attention mechanisms to the graph neural network field Veličković et al. [2017]. This innovation effectively addressed limitations of traditional methods in handling neighbor importance differences, opening a new paradigm of dynamic weight learning Veličković et al. [2017].

Graph Convolutional Networks Graph Convolutional Networks (GCN) represent a milestone in modern graph neural networks, with theoretical foundations deeply rooted in spectral graph theory Kipf and Welling [2016]. GCN cleverly extends traditional convolution operations to graph-structured data by defining convolutions in the graph’s frequency domain to achieve feature learning Kipf and Welling [2016]. Specifically, the convolution of graph signal x with filter g_θ is defined as $g_\theta * x = Ug_\theta(\Lambda)U^Tx$, where U and Λ are the eigenvector matrix and eigenvalue matrix of the graph Laplacian, respectively Kipf and Welling [2016].

To avoid expensive eigendecomposition computations, Kipf and Welling proposed a simplified GCN model using first-order approximation of Chebyshev polynomials, simplifying the graph convolution operation to $H^{(l+1)} = \sigma(\tilde{A}H^{(l)}W^{(l)})$, where $\tilde{A} = D^{-1/2}AD^{-1/2}$ is the symmetric normalized adjacency matrix Kipf and Welling [2016]. This design achieves linear computational complexity $O(|E|)$, enabling GCN to scale to large-scale graph data processing Kipf and Welling [2016].

The main advantages of GCN include high computational efficiency, effective utilization of local structural information in graphs, and solid theoretical foundations Kipf and Welling [2016]. Its core idea is to update current node representations by aggregating feature information from neighbor nodes, thereby capturing structural characteristics of graphs Kipf and Welling [2016]. However, GCN also has widely recognized limitations: its aggregation weights are predefined by the graph’s Laplacian matrix, making it limited when processing weighted graphs or graphs requiring dynamic adjustment of neighbor importance Kipf and Welling [2016]. Additionally, GCN primarily learns low-pass filters and may perform poorly on tasks requiring high-frequency information Kipf and Welling [2016].

Graph Attention Networks Graph Attention Networks (GAT) were proposed to address the fundamental limitations of GCN’s fixed aggregation strategy Veličković et al. [2017]. Traditional GCN treats all neighbor nodes equally, unable to distinguish the importance of different neighbors, which appears inadequate when processing complex graph structures Veličković et al. [2017]. The core innovation of GAT lies in introducing self-attention mechanisms, allowing models to dynamically learn importance weights of neighbor nodes, thereby achieving more flexible feature aggregation Veličković et al. [2017].

Technically, GAT first computes attention coefficients between node i and its neighbor node j as $e_{ij} = a^T[Wh_i||Wh_j]$, where W is a learnable weight matrix, a is the attention vector, and $||$ represents vector concatenation Veličković et al. [2017]. After processing attention coefficients through LeakyReLU activation function, soft-

max normalization yields $\alpha_{ij} = \text{softmax}_j(\text{LeakyReLU}(e_{ij}))$ Veličković et al. [2017]. The final node representation is obtained through weighted aggregation of neighbor features: $h'_i = \sigma(\sum_{j \in N(i)} \alpha_{ij} W h_j)$ Veličković et al. [2017].

The fundamental difference between GAT and GCN lies in the flexibility of aggregation strategies: GCN uses fixed normalized weights while GAT learns dynamic weights through attention mechanisms Veličković et al. [2017]. This gives GAT significant advantages: stronger expressive capability for capturing more complex node relationships; no need for prior knowledge of complete graph structure, with stronger inductive learning capability; more flexible processing of nodes with different degrees, unaffected by graph structure changes Veličković et al. [2017].

Comparative Analysis Framework From a comparative analysis perspective, GCN and GAT represent two different design philosophies in graph neural networks Veličković et al. [2017], Kipf and Welling [2016]. GCN employs fixed-weight Laplacian normalized aggregation with computational complexity $O(|E|)$, relatively fewer parameters, solid theoretical foundations based on spectral graph theory, but limited inductive capability requiring fixed graph structure Kipf and Welling [2016]. In contrast, GAT employs dynamic attention weight aggregation with the same computational complexity $O(|E|)$, but more total parameters due to attention parameters, theoretical foundations based on attention mechanisms, and stronger inductive capability and adaptability Veličković et al. [2017].

Regarding application scenarios, GCN is suitable for scenarios with relatively stable graph structures and relatively equal importance of node relationships, while GAT is more appropriate for complex applications requiring differentiation of neighbor importance, dynamic graph structure changes, or heterogeneous characteristics Veličković et al. [2017], Kipf and Welling [2016]. Both methods perform excellently in tasks such as node classification, graph classification, and link prediction, but GAT typically demonstrates better adaptability and robustness when processing complex graph structures Veličković et al. [2017].

Through this comparative analysis, we can clearly see that GCN and GAT represent fixed aggregation and dynamic aggregation design philosophies respectively Veličković et al. [2017], Kipf and Welling [2016]. GAT, by introducing attention mechanisms, provides more flexibility than GCN in handling neighbor importance, establishing a solid theoretical foundation for subsequent method selection and comparative analysis Veličković et al. [2017].

2.3.2 3D Convolutional Neural Networks

Traditional 3D CNN Architectures

Traditional convolutional neural network (CNN) architectures typically employ a sequence of convolutional layers followed by activation functions, such as Rectified Linear Unit (ReLU), in the pattern of Conv $\rightarrow$ Activation $\rightarrow$ Conv $\rightarrow$ Activation. However, an alternative approach, where two consecutive convolutional layers are

applied before activation ($\text{Conv} \rightarrow \text{Conv} \rightarrow \text{Activation}$), has demonstrated several advantages. This section explores the theoretical underpinnings and empirical evidence supporting this structure, which has been effectively utilized in architectures such as ResNet, Inception, and U-Net.

Double Convolution: An Alternative Design Paradigm

Advantages of Delaying Activation in DoubleConvs Preserving More Linear Combinations and Reducing Information Loss In conventional CNNs, applying ReLU immediately after each convolution introduces non-linearity at an early stage. While this can enhance representational capacity, it also results in the premature loss of critical feature information by setting negative values to zero. In contrast, by deferring activation until after two convolutional layers, the network ensures that:

- The first convolution extracts fundamental low-level features such as edges and textures.
- The second convolution refines and enriches these extracted features before activation.
- Non-linearity is applied only after meaningful representations have been developed, preserving useful information that may otherwise be discarded prematurely.

This approach is supported by the deep residual learning framework, where [He et al. \[2016a\]](#) demonstrated that delaying non-linearity contributes to improved feature retention and learning efficiency.

Enhancing Feature Complementarity for Better Representation

Stacking two convolutional layers before activation allows for more effective feature learning. The first convolution acts as a basic feature extractor, capturing elementary structures, while the second convolution combines and refines these features into more complex representations. This design is particularly beneficial in deep networks, where hierarchical features are crucial for accurate predictions.

[Szegedy et al. \[2015\]](#) leveraged a similar principle in the Inception-v3 architecture, where non-linearity was selectively delayed in certain blocks to facilitate improved feature aggregation. By postponing activation, the network enhances spatial coherence and strengthens complementary feature representations before enforcing non-linearity.

Reducing Gradient Vanishing Issues for Smoother Backpropagation

Overuse of activation functions such as ReLU in deep networks can contribute to gradient vanishing issues, as excessive non-linearity disrupts smooth information flow. By delaying activation, the network:

- Mitigates excessive early-stage ReLU-induced gradient clipping.

- Maintains stable gradients, especially in deep architectures.
- Facilitates more effective weight updates during backpropagation.

Glorot et al. [2011] highlighted the potential drawbacks of aggressive early-stage activation, noting that while ReLU addresses vanishing gradients to some extent, its overuse may hinder optimization. The DoubleConv strategy provides a balance, allowing for richer feature propagation before non-linearity is introduced.

Improved Performance in Semantic Segmentation Tasks

Semantic segmentation models, such as U-Net and DeepLabV3+, often require fine-grained spatial information to achieve precise boundary delineation. Applying activation prematurely can lead to:

- Over-pruning of features, resulting in loss of spatial details.
- Poor edge detection, reducing segmentation accuracy.

By adopting the Conv $\rightarrow$ Conv $\rightarrow$ Activation approach, segmentation networks retain more spatial information before enforcing non-linearity, leading to smoother segmentations and improved edge preservation. Ronneberger et al. [2015a] demonstrated the efficacy of this approach in U-Net, where it significantly enhanced biomedical image segmentation by preventing early-stage feature loss.

Limitations of DoubleConv While DoubleConv offers several advantages, it also has notable drawbacks. One major limitation is the increased computational cost, as the addition of an extra convolutional layer increases the number of computations per forward pass. This can lead to higher latency during inference, which is particularly problematic in real-time applications such as edge AI and mobile vision processing. Furthermore, the increased computational demand makes DoubleConv less suitable for embedded systems or devices with strict hardware constraints Howard et al. [2017].

Another drawback is the potential risk of overfitting, especially when training on small datasets. Since DoubleConv introduces more trainable parameters, it increases the capacity of the model to learn complex patterns, but this can also lead to overfitting if there is insufficient data to generalize well. In scenarios where model generalization is critical, such as medical imaging with limited labeled samples, alternative regularization techniques or simpler architectures may be preferable Srivastava et al. [2014b].

In shallow CNN architectures or simple classification tasks, the additional convolution layer may be redundant. If the first convolution already captures sufficient meaningful features, adding another convolution might not provide significant improvements and could instead introduce unnecessary computational overhead. Early CNN models, such as LeNet-5, successfully used a single convolution per block to achieve strong performance in basic recognition tasks LeCun et al. [1998]. Thus, for lightweight models designed for simple image classification, the benefits of DoubleConv may not justify the additional complexity.

Subgroup	Elements	Symmetry Body	Key Property	Compute Cost
T_4	12	tetrahedron	even permutations limited coverage	low
S_4	24	cube	full 90° cube rotations non-abelian	$\approx 2 \times T_4$

Rotation-Invariant Extensions

Traditional convolutional neural networks (CNNs) are inherently equivariant to translations, but are not equivariant to rotations. In the context of 3D voxel-based medical images, where structures of interest can appear in various orientations, this limitation can reduce generalization ability unless rotation-based data augmentation is applied extensively. Group Equivariant CNNs provide a theoretical solution to this limitation.

SO(3) Equivariant Convolutions via Discrete Approximation CubeNet ? addresses this issue by incorporating *group equivariant convolution* to achieve equivariance to a discrete set of 3D right-angle rotations. The key idea is to design convolutional layers that respect a predefined symmetry group G , such that for any transformation $g \in G$, the feature transformation satisfies:

$$\Phi(gx) = g\Phi(x) \quad \forall g \in G$$

where G is the chosen subgroup and Φ denotes the feature mapping. To make computation tractable, the continuous rotation group $SO(3)$ is discretized into finite subgroups: Klein’s four-group V_4 (4 elements), the tetrahedral group T_4 (12 elements), and the cube group S_4 (24 elements).

Comparing T_4 and S_4 In highly symmetric, axis-aligned medical volumes (CT, MRI), S_4 roughly doubles the computation but offers more complete directional coverage, often yielding better rotational generalization.

Relevance to Medical Voxel Data Medical scans are usually acquired in standardized orientations, yet small patient rotations or scanner misalignments persist. Employing S_4 -equivariant convolutions allows the network to accommodate such variations without exhaustive data augmentation, preserving voxel-grid alignment and reducing preprocessing overhead.

2.3.3 Feature Compression and Representation Learning

Mathematical Analysis of the Curse of Dimensionality

High-dimensional feature processing in medical imaging faces fundamental challenges rooted in the curse of dimensionality. As the dimensionality of feature spaces increases, several critical problems emerge that directly impact the effectiveness of machine learning models.

Exponential Growth of Parameter Space In traditional MLP-based approaches, the parameter count grows exponentially with input dimensionality. For a feature tensor with dimensions $D_1 \times D_2 \times \dots \times D_n$, the flattened representation requires $\prod_{i=1}^n D_i$ parameters for the first linear layer alone. This exponential growth leads to several critical issues:

- **Memory Constraints:** Modern GPUs have limited memory capacity, and storing millions or billions of parameters becomes computationally prohibitive.
- **Training Instability:** Large parameter spaces are prone to vanishing and exploding gradients, making optimization challenging.
- **Overfitting Risk:** With limited medical imaging datasets, the ratio of parameters to training samples becomes unfavorable, leading to poor generalization.

Sparsity and Distance Concentration In high-dimensional spaces, data points become increasingly sparse, and the concept of distance loses its discriminative power. This phenomenon, known as distance concentration, means that the ratio of the farthest to nearest neighbor approaches 1 as dimensionality increases:

$$\lim_{d \rightarrow \infty} \frac{\text{dist}_{\max} - \text{dist}_{\min}}{\text{dist}_{\min}} \rightarrow 0$$

This has profound implications for medical imaging applications where feature similarity is crucial for accurate diagnosis and classification.

MLP vs NdLinear Theoretical Comparison

In modern medical imaging analysis, deep learning models, particularly Convolutional Neural Networks (CNNs), have become indispensable tools. These models can automatically extract semantically rich features from high-dimensional data such as Magnetic Resonance Imaging (MRI) or Computed Tomography (CT) scans for tasks like disease classification, lesion segmentation, or prognostic prediction [Wang et al., Ker et al. \[2018\]](#). However, after a deep encoder processes a 3D or even 4D medical image, it typically produces a high-dimensional feature tensor. A critical challenge lies in effectively compressing these features and projecting them into a lower-dimensional space for final decision-making.

Traditional Method: MLP-Based Feature Compression Theory An MLP-based projection head is one of the most common designs in deep learning. Traditionally, the most direct approach involves "flattening" the high-dimensional tensor into a one-dimensional vector, which is then fed into a Multi-Layer Perceptron (MLP), often referred to as a "projection head." This practice, however, has significant theoretical and practical limitations. As noted by [Reneau et al. \[2025\]](#), this "ubiquitous flattening, even if localized, is often detrimental. It discards the data's explicit N-D organization, disrupting critical spatial, temporal, or other relationships encoded across dimensions."

Theoretical Limitations of MLP

Although the MLP possesses the powerful capability of a "universal approximator," its inherent flattening operation introduces severe problems when processing high-dimensional structured data:

- **Destruction of Structural Information:** The value of medical images lies in the continuity of their spatial structure and local correlations. For instance, in a 3D MRI scan, the relationship between a voxel and its surrounding voxels is crucial for determining tissue pathology. The flattening operation completely disrupts these three-dimensional spatial adjacencies. This loss of explicit structure forces the network to relearn intrinsic dependencies from a decontextualized vector, which increases learning difficulty and risks information loss [Reneau et al. \[2025\]](#).
- **Parameter Inefficiency (Curse of Dimensionality):** The number of parameters in an MLP grows exponentially with the dimensions of its input and output. For a high-dimensional medical image feature tensor, the length of the flattened 1D vector becomes immense, leading to an explosion in the parameter count of the first fully-connected layer. This not only creates a substantial computational and memory burden but also poses a significant challenge in the medical imaging domain, where datasets are often limited in size. An excess of parameters can easily lead to model overfitting, thereby reducing its generalization ability, a phenomenon also discussed in the context of dimension-reduction techniques for MLPs [Yang et al. \[2020\]](#).

MLP Parameter Complexity Analysis For an MLP processing a flattened tensor of size $N = \prod_{i=1}^d D_i$, the parameter count for a single hidden layer of size H is:

$$P_{\text{MLP}} = N \times H + H = H(N + 1)$$

For multiple layers, this grows as:

$$P_{\text{MLP, total}} = \sum_{l=1}^L H_l(H_{l-1} + 1)$$

where $H_0 = N$ is the input size.

Emerging Method: NdLinear-Based Feature Compression Theory Recently, a novel linear layer named NdLinear has been proposed, designed to operate directly on N-dimensional tensors without flattening, offering a new paradigm for high-dimensional feature compression [Reneau et al. \[2025\]](#). To address the problems introduced by the flattening operation, [Reneau et al. \[2025\]](#) proposed NdLinear, a novel linear layer that operates directly on N-dimensional tensors without the need for flattening.

Core Mechanism and Theoretical Advantages of NdLinear

The core idea of NdLinear is "dimension-wise transformation." Instead of treating the N-D tensor as a monolithic block, it designs an independent weight matrix for each dimension (or "mode") and applies a linear transformation sequentially along each one. This design offers several significant theoretical advantages:

- **Preservation of Native N-D Structure:** This is the most fundamental advantage of NdLinear. By operating directly on the original multi-dimensional space, the structural information across dimensions in medical images—such as spatial (width, height, depth), channel, or even temporal (for 4D data)—is fully preserved. This allows the model to learn cross-dimensional feature correlations more naturally and effectively.
- **Extreme Parameter Efficiency:** In contrast to the exponential parameter growth in MLPs ($\prod D_k \times \prod H_k$), the parameter count of NdLinear is merely the sum of its dimension-wise transformations ($\sum_{k=1}^N (D_k H_k + H_k)$). This translates to a parameter reduction that can be orders of magnitude.
- **Inherent Inductive Bias and Regularization Effect:** The architecture of NdLinear itself serves as a powerful inductive bias, as it assumes that features can be processed separably along each dimension. This structural constraint naturally acts as a form of regularization, mitigating the risk of overfitting on smaller datasets.

NdLinear Parameter Efficiency Analysis NdLinear processes each dimension independently, resulting in a parameter count of:

$$P_{\text{NdLinear}} = \sum_{i=1}^d (D_i \times H_i + H_i) = \sum_{i=1}^d H_i (D_i + 1)$$

The parameter efficiency ratio between NdLinear and MLP is:

$$R = \frac{P_{\text{NdLinear}}}{P_{\text{MLP}}} = \frac{\sum_{i=1}^d H_i (D_i + 1)}{H(\prod_{i=1}^d D_i + 1)}$$

For typical medical imaging scenarios where $D_i \approx D$ for all dimensions, this ratio becomes:

$$R \approx \frac{d \cdot H \cdot (D + 1)}{H \cdot (D^d + 1)} \approx \frac{d(D + 1)}{D^d}$$

This demonstrates that NdLinear achieves exponential parameter reduction as dimensionality increases. For instance, [Reneau et al. \[2025\]](#) illustrate that for a 3D tensor transformation where input and output dimensions are all 32, a standard linear layer would require approximately $32^6 \approx 10^9$ parameters, whereas NdLinear would only need $3 \times 32^2 \approx 3 \times 10^3$. This property is critically important for medical imaging tasks, which often contend with limited samples and high-dimensional features.

Theoretical Advantages of Structural Preservation

Structural Preservation in Medical Imaging Medical images possess inherent structural relationships across spatial, temporal, and channel dimensions. NdLinear preserves these relationships by maintaining the tensor structure throughout processing, enabling the model to:

- Capture spatial correlations in 3D anatomical structures
- Preserve temporal dynamics in 4D medical sequences
- Maintain channel relationships in multi-modal imaging

Regularization Through Architecture The dimension-wise processing in NdLinear introduces an implicit regularization effect. By constraining transformations to operate along individual dimensions, the architecture enforces a separability assumption that acts as a strong inductive bias. This is particularly beneficial for medical imaging where:

- Anatomical structures often exhibit separable patterns across different spatial dimensions
- Multi-modal data can be processed independently while preserving cross-modal relationships
- Limited training data benefits from architectural constraints that reduce overfitting

Computational Efficiency Theoretical Foundation Beyond parameter efficiency, NdLinear offers computational advantages:

- **Memory Efficiency:** Lower parameter count reduces memory requirements during training and inference

- **Parallel Processing:** Dimension-wise operations can be parallelized across different dimensions
- **Gradient Flow:** Shorter computational paths improve gradient propagation and training stability

2.3.4 Loss Functions and Optimization

L1 vs MSE Loss Theoretical Analysis

In the training of Autoencoders, the choice of loss function is critical to the quality of the reconstructed output. Among the most fundamental and widely used reconstruction losses are Mean Squared Error (MSE) and L1 loss (Mean Absolute Error, MAE). While MSE is popular due to its mathematical simplicity and ease of optimization, its inherent limitations become particularly detrimental when applied to 3D medical imaging. As a common alternative, L1 loss offers improvements in some respects; however, as both are pixel-wise (or voxel-wise) metrics, they share common challenges in preserving the fine structures and topological information essential for clinical diagnosis.

Comparative Analysis of Mathematical Properties The core task of a reconstruction loss in an autoencoder is to quantify the discrepancy between the original and reconstructed images. The fundamental difference between how MSE and L1 loss perform this task directly leads to different outcomes in visual clarity.

The Blurring Effect of MSE: The core mechanism of MSE involves calculating the **square** of the difference between the original and reconstructed outputs for each voxel. This incentivizes the model to find a "smooth" solution that minimizes the overall average error. While this facilitates rapid convergence, the trade-off in medical imaging is the sacrifice of high-frequency details, such as sharp bone boundaries, soft tissue textures, or subtle features of lesions. This loss of detail results in blurry reconstructions, which can compromise subsequent clinical interpretation or automated analysis tasks. Research by [Valanarasu et al. \[2024\]](#) explicitly highlighted this deficiency in standard Masked Autoencoders (which typically rely on MSE for reconstruction), observing that such models fail to effectively reconstruct bones and other fine structures.

Improved Clarity and Inherent Limitations of L1 Loss: Unlike MSE, L1 loss computes the **absolute** difference between the original and reconstructed values. Because its penalty for errors is linear, it does not disproportionately penalize larger deviations. Consequently, it is more inclined to preserve edges and details during reconstruction, leading to sharper images. For this reason, many studies aiming for higher reconstruction quality opt for L1 loss as the primary objective.

However, even with its superior performance in clarity, L1 loss has its own limitations. In a study on MR-to-CT image translation, [Boutillon et al. \[2024\]](#) noted that even L1 loss, as a global function, primarily measures overall intensity differences

without adequately considering the preservation of anatomical structures. They argued that such losses tend to smooth over finer details, resulting in images that may appear realistic at a macroscopic level but lack the precision required for clinical tasks. The study further advised against using L1 loss for tasks requiring the generation of fine structures like vessels or bronchi, as it still has a tendency to produce blurry reconstructions that can obscure important details. This demonstrates that while L1 is an improvement over MSE, it remains an imperfect solution for highly complex medical images.

Geometric and Topological Information Limitations Inability to Capture Geometric and Topological Information Organs and tissues in 3D medical images are defined not only by their voxel intensities but, more importantly, by their geometric shapes and topological structures in three-dimensional space, such as connectivity and cavities. Both MSE and L1 loss, being voxel-wise metrics, are inherently blind to this macroscopic structural information. A model may perform well according to L1 or MSE metrics, yet the reconstructed organ could exhibit structural errors, such as spurious disconnections or incorrect connections.

To address this issue, the research community has proposed solutions that go beyond pixel-level comparisons. [Gu et al. \[2024\]](#) argued that standard reconstruction tasks are insufficient for capturing the geometric and spatial information critical for medical image segmentation, thus motivating the introduction of a "Topological Loss." They stated that existing pre-training methods based on Masked Autoencoders lack the ability to capture geometric shape and spatial information, which is crucial for medical image segmentation tasks. The very motivation for this research provides a powerful argument: if standard reconstruction with L1 or MSE were sufficient, there would be no need to develop a specialized topological loss. Their experimental results confirmed that relying solely on pixel-wise losses leads to an inadequate understanding of geometric structures by the model.

Hamming Loss Information-Theoretic Foundation

The Hamming Loss is a key metric for evaluating the performance of machine learning classification models, especially in the context of multi-label classification [Wikipedia \[2023\]](#). Its core concept originates from the Hamming Distance in information theory, initially developed by Richard Hamming for detecting and correcting errors in data transmission. In machine learning applications, it has been adapted into a loss function that measures the model's prediction accuracy by calculating the fraction of incorrectly predicted labels relative to the total number of labels [Scikit-learn developers \[2024\]](#), [Number Analytics \[2025\]](#).

Mathematical Definition and Properties The value of the Hamming Loss ranges between 0 and 1, where a lower value indicates a more accurate prediction. A Hamming Loss of 0 signifies that the model has perfectly predicted all labels for all samples [Hüllermeier et al. \[2020\]](#).

The Hamming Loss is calculated by dividing the total number of misclassified labels by the product of the total number of samples and the total number of labels:

$$\text{HammingLoss} = \frac{1}{N \times L} \sum_{i=1}^N \sum_{j=1}^L \mathbb{I}(y_{i,j} \neq p_{i,j})$$

where N represents the total number of samples, L represents the total number of classes (labels), $y_{i,j}$ represents the true value of the j -th label for the i -th sample, and $p_{i,j}$ represents the predicted value. The indicator function $\mathbb{I}(y_{i,j} \neq p_{i,j})$ returns 1 if the true and predicted values differ and 0 otherwise.

Theoretical Advantages in Multi-Label Classification In cellular matching tasks, where each cell correspondence can be viewed as a binary classification problem, Hamming Loss provides several theoretical advantages:

Provides Granular Error Analysis: Unlike subset accuracy metrics, Hamming Loss penalizes each incorrectly predicted correspondence individually, offering fine-grained analysis of matching errors.

Intuitive and Easy to Interpret: The metric directly reflects the average fraction of correspondences that were incorrectly predicted per sample.

Treats All Correspondences Equally: This metric assigns equal weight to each matching decision, which is appropriate for cellular matching where all correspondences are equally important.

Combinatorial Optimization Theory

Hungarian Algorithm Mathematical Foundation The Hungarian algorithm provides the theoretical foundation for solving the Linear Assignment Problem (LAP) in polynomial time with complexity $O(n^3)$. For cellular matching applications, the algorithm finds the optimal one-to-one correspondence between cells in two different organisms by minimizing the total matching cost.

Differentiable Optimization Principles The integration of combinatorial solvers into neural networks requires differentiable approximations of discrete optimization processes. The perturbation-based gradient estimation technique enables backpropagation through combinatorial layers by approximating gradients through small perturbations of the input cost matrix.

Hamming Loss for Matching Evaluation

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into a loss function that measures the model’s prediction accuracy by calculating the fraction of incorrectly predicted labels relative to the total number of labels [Scikit-learn developers \[2024\]](#), [Number Analytics \[2025\]](#).

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The Hamming Loss is calculated by dividing the total number of misclassified labels by the product of the total number of samples and the total number of labels:

$$\text{HammingLoss} = \frac{1}{N \times L} \sum_{i=1}^N \sum_{j=1}^L \mathbb{I}(y_{i,j} \neq p_{i,j})$$

where N represents the total number of samples, L represents the total number of classes (labels), $y_{i,j}$ represents the true value of the j -th label for the i -th sample, and $p_{i,j}$ represents the predicted value. The indicator function $\mathbb{I}(y_{i,j} \neq p_{i,j})$ returns 1 if the true and predicted values differ and 0 otherwise.

For cellular matching applications, this translates to evaluating the accuracy of cell correspondence predictions across multiple organism pairs, providing a robust metric for assessing matching performance that accounts for both correct and incorrect assignments in a balanced manner.

The integration of these diverse loss functions and optimization strategies provides the theoretical foundation necessary for developing robust cellular matching algorithms that can effectively balance reconstruction quality, perceptual fidelity, and matching accuracy in complex 3D microscopy environments.

3 Method

3.1 Proposed Method Framework

3.1.1 System Architecture Overview

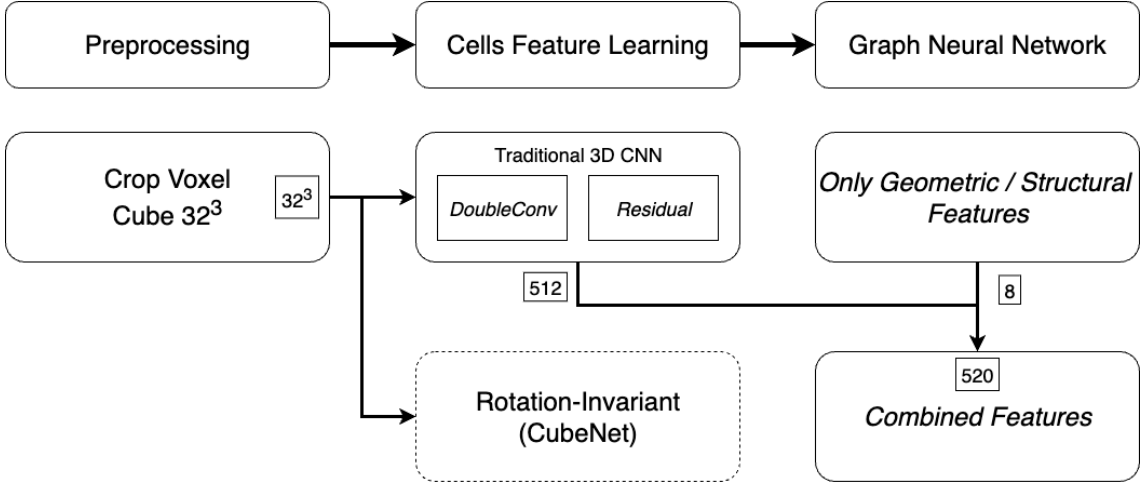


Figure 3.1: Overview of the proposed deep learning architecture for cell matching. The system processes 32^3 voxel subvolumes through a feature learning module (traditional 3D CNN or rotation-invariant CubeNet), followed by graph neural network processing for context-aware refinement. Extracted feature dimensions are indicated at each processing stage ($512+8 \rightarrow 520$). The dashed box represents an alternative rotation-invariant approach that showed limited effectiveness in our experiments because an alternative rotation-invariant approach whose full potential could not be fully evaluated due to the computational and implementation constraints detailed in Section 3.2.4.

The proposed architecture consists of four main processing stages:

1. **Preprocessing Module:** Extracts 32^3 voxel subvolumes centered at annotated cell coordinates to serve as standardized input patches.
2. **Feature Learning Module:** Employs either traditional 3D CNN architecture with DoubleConv and Residual blocks, or an alternative rotation-invariant CubeNet approach for spatial feature extraction.

3. **Graph Processing Module:** Constructs spatial graphs based on geometric relationships between cells and applies graph attention networks (GAT) to enable context-aware feature refinement through message passing. The main metrics we consider is based on **Feature Integration** which combines geometric and structural features (8 dimensions) with learned CNN representations to produce final combined features (520 dimensions).

The effectiveness of each italicized component is systematically evaluated through comprehensive ablation studies presented in Experiments chapter.

3.1.2 Processing Pipeline Overview

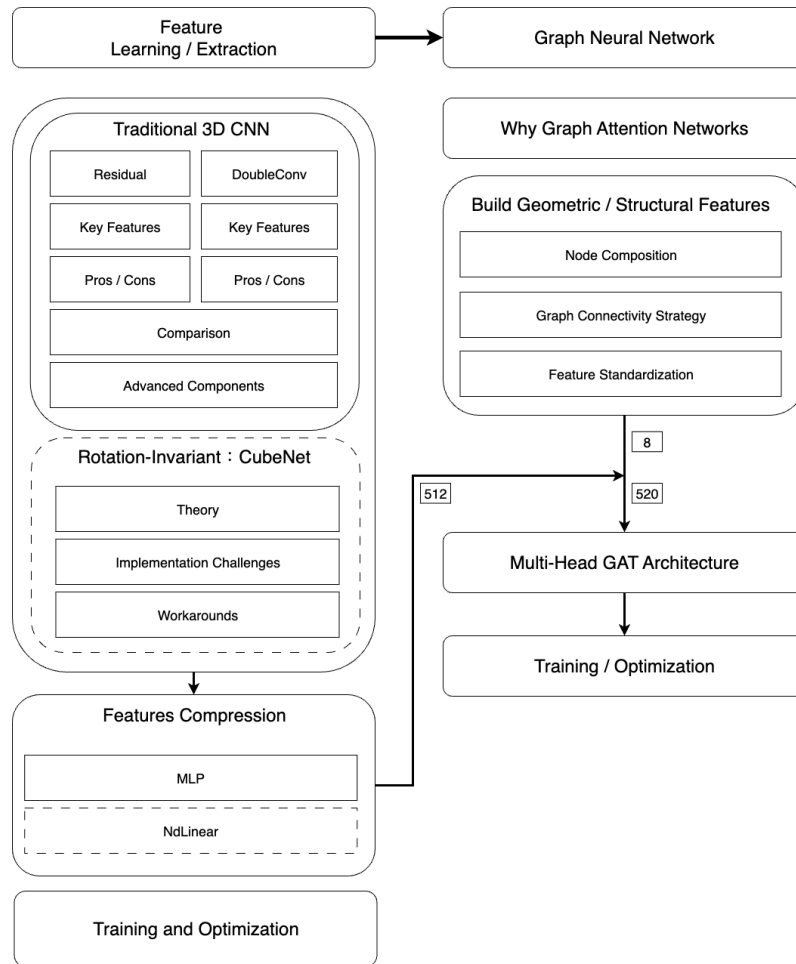


Figure 3.2: The proposed processing pipeline for cell matching. The workflow consists of two main stages: (1) feature extraction and compression using 3D CNN-based methods, and (2) graph construction and matching using a Graph Attention Network (GAT).

Figure 3.2 presents the comprehensive processing pipeline designed for our cell matching task. The pipeline is structured into two sequential stages: Feature Learning and Graph-based Matching.

In the first stage, Feature Learning and Extraction, we explore two distinct 3D Convolutional Neural Network (3D CNN) approaches. The primary method is a traditional 3D CNN architecture. As an alternative, we investigate a rotation-invariant model, CubeNet. As indicated by the dashed box, CubeNet presents implementation challenges that are not yet fully resolved; these challenges and our attempted workarounds are documented as part of our methodology. Following feature extraction, a Feature Compression step is applied. This is crucial because Graph Attention Networks (GATs) operate more effectively on lower-dimensional inputs (e.g., under 1024 dimensions). We primarily employ a Multi-Layer Perceptron (MLP), derived from a trained autoencoder’s encoder, for this purpose. NdLinear is noted as a potential alternative, though it requires further validation.

The second stage focuses on the Graph Neural Network. We selected the Graph Attention Network (GAT) for its ability to weigh the importance of different nodes in a neighborhood. The process begins with building a graph using the cells’ geometric and structural properties to define node composition and connectivity strategies. Finally, the compressed features from the first stage are integrated as node attributes. This composite graph is then fed into a Multi-Head GAT architecture for end-to-end training and optimization to evaluate the final matching performance.

3.2 Feature Learning and Extraction

3.2.1 Design Rationale and Considerations

According to findings by [Bondarenko et al. \[2023\]](#) in embryo-uterine interaction studies, 3D microscopy imaging had unique characteristics that provided important guidance for 3D CNN design. Anisotropic resolution was a key feature, where Z-axis resolution was typically lower than X-Y axes, requiring special convolution kernel designs to adapt to resolution differences across different axes. This anisotropic characteristic was prevalent in various 3D microscopy imaging techniques, including Light Sheet Fluorescence Microscopy (LSFM) and confocal microscopy.

The choice of 32^3 voxel cube size was determined based on cellular morphology analysis, where this dimension provided approximately 4 additional voxels beyond the typical cell boundary, ensuring comprehensive capture of cellular features while maintaining computational efficiency.

In [Bondarenko et al. \[2023\]](#)’s study, LSFM images had PSNR of 22.5 ± 6.5 dB, while confocal images had 25.9 ± 5.7 dB, requiring networks with good noise resistance capabilities. Additionally, cell density variations existed significantly across different tissue regions, requiring networks capable of adapting to multi-scale cell distribution patterns. Notably, similar imaging characteristics were observed in PlantSeg studies, where light sheet microscopy datasets had average PSNR of 22.5 ± 6.5 dB, 3.4 dB

lower than confocal datasets, similarly reflecting the anisotropic feature where Z-axis resolution was inferior to the other two axes Wolny et al. [2020a].

3.2.2 Architecture Design Principles

Based on successful experiences from PlantSeg research, 3D CNNs demonstrated excellent performance in biological image segmentation Wolny et al. [2020a]. To address the anisotropic characteristics prevalent in 3D microscopy imaging, network architecture design had to specifically consider the problem of lower Z-axis resolution. PlantSeg adopted innovative anisotropic 3D designs: using 2D convolutions at the finest scale to handle regions with maximum anisotropy, while using combinations of $3\times3\times1$ and $3\times3\times3$ convolutions at other scales, with each module representing $7\times7\times5$ nonlinear computations showing slight anisotropic features.

In terms of network architecture selection, symmetric encoder-decoder structures of 3D U-Net variants were adopted, particularly suitable for dense prediction tasks. The motivation for architectural design centered on two distinct approaches to feature extraction: **residual-based** and **sequential convolution-based** paradigms, each offering unique advantages for cellular feature representation.

The breakthrough achievements by Lee et al. [2017] in the SNEMI3D connectomics challenge provided important design guidance for residual-based approaches. This study similarly faced high anisotropic characteristics of serial section EM images, adopting targeted architectural designs to handle the problem where Z-axis resolution was inferior to X-Y axes. The method adopted a residual symmetric U-Net architecture, specifically using 2D convolutions at the finest scale (top level) to handle regions with maximum anisotropy, while other scales used combinations of $3\times3\times1$ and $3\times3\times3$ convolutions. This design was the first to surpass human accuracy estimates on the SNEMI3D benchmark, reducing Rand error to 0.02576, fully demonstrating the effectiveness of 3D CNN architectures designed for anisotropic characteristics.

3.2.3 Normal 3D CNN Feature Extraction

Residual Block Variant

The feature extraction stage involved deep feature learning, adopting 3D CNN designs based on PlantSeg Wolny et al. [2020a], specifically optimized for anisotropic characteristics of microscopy imaging. The overall architecture adopted an Autoencoder design, where the encoder part was responsible for compressing 32^3 input image blocks with feature channels into 512-dimensional high-dimensional feature vectors. Each module added residual skip connections, effectively improving gradient flow He et al. [2016a]. The motivation for this design was to first embed 2D features, then refine them through residuals from 3D context, fully adapting to the high anisotropic characteristics of serial section images Lee et al. [2017].

ResNetBlock: Innovations and Comparisons with Double Convolution

Even though our data consists of fluorescence microscopy images, machine learning models developed for electron microscopy (EM) still maintain a strong relationship with our type of data. The work by Lee et al. [2017] represents a significant breakthrough, as their proposed approach outperformed human experts in segmentation accuracy. This section discusses their key contributions, limitations, and how their method compares with the traditional Double Convolution (DoubleConv) approach commonly used in U-Net-based architectures.

Key Architectural Innovations The innovations introduced by Lee et al. [2017] contributed substantially to their model’s unprecedented accuracy in SNEMI3D. The adoption of a Residual Symmetric U-Net incorporated 3D convolutions in deeper layers while retaining 2D convolutions in finer resolution levels. This hybrid design effectively addressed the anisotropic nature of EM images, ensuring that Z-axis information was not lost while preserving critical high-resolution features. To mitigate common merge errors in segmentation, the model introduced a novel long-range affinity prediction mechanism. Unlike traditional affinity maps that predict nearest-neighbor relationships, their method extended affinity prediction to 3, 9, and 27 voxel distances. This enhancement improved the network’s ability to capture contextual dependencies across large regions, thereby improving segmentation consistency.

Computational Efficiency and Performance Optimization Instead of relying on computationally expensive Test-Time Augmentation (TTA), the study introduced Mean Affinity Agglomeration, a novel postprocessing method that improved segmentation accuracy while significantly reducing inference time Lee et al. [2017]. This method smooths affinity scores across supervoxels, effectively reducing noise and preventing spurious merges. Despite surpassing human accuracy, the model maintained a relatively small parameter size of 1.5M parameters compared to alternative models like DIVE ($18M \times 3$ models) and IAL (35M parameters). Moreover, it eliminated the need for TTA, further accelerating inference.

Generalization and Dataset Limitations Despite its impressive performance, the proposed method faces several limitations. The primary concern relates to its generalization capability, as the model was trained and evaluated on the SNEMI3D dataset, which represents a relatively narrow scope of electron microscopy images. The effectiveness on more diverse and larger-scale datasets remains uncertain, necessitating broader validation across different EM datasets to confirm its robustness in real-world applications.

Persistent Segmentation Challenges The presence of persistent segmentation errors presents another challenge. While the model significantly improved accuracy, it still struggles with very thin neurites that are prone to missegmentation, as well

as self-touching structures such as dendritic spines, which can be incorrectly merged with their parent structures. Additionally, certain underrepresented cell types, such as glial cells, remain difficult to classify accurately due to limited training samples Lee et al. [2017].

Computational Resource Requirements The computational demands of the method also pose limitations. Training the model requires between 500K and 700K iterations and takes approximately five days on a high-performance GPU, making it an expensive process. While the model improves inference speed by eliminating TTA, its training cost remains substantial, which could hinder accessibility and deployment in research environments with limited computational resources Lee et al. [2017].

Comparison with Double Convolution (DoubleConv) While DoubleConv has been widely used in U-Net architectures, particularly for biomedical image segmentation, the approach introduced by Ronneberger et al. [2015b] provides several distinct advantages in processing electron microscopy (EM) images. One of the primary improvements is in handling anisotropic EM images. Traditional DoubleConv structures typically rely on two consecutive 3×3 convolutions, which may not fully leverage the anisotropic nature of EM images. In contrast, Ronneberger et al. [2015b]’s method integrates 2D convolutions in high-resolution layers with 3D convolutions in deeper layers. This hybrid design ensures that crucial information along the Z-axis is preserved while simultaneously reducing computational overhead.

Another key distinction lies in the ability to capture long-range dependencies. DoubleConv mainly focuses on local feature extraction and does not inherently model long-range spatial relationships. On the other hand, Ronneberger et al. [2015b]’s method introduces long-range affinity prediction, allowing the network to learn neuronal structures across larger spatial distances. This approach significantly improves segmentation integrity and reduces the likelihood of merge errors, which are common in neuronal reconstruction tasks.

From a computational perspective, DoubleConv architectures are often considered lightweight, but they may suffer from information loss due to immediate non-linearity application. Despite incorporating 3D convolutions, Ronneberger et al. [2015b]’s approach remains computationally efficient by maintaining a smaller parameter count and eliminating the reliance on TTA. This combination allows their model to achieve superior segmentation accuracy without excessively increasing computational complexity.

Alternative Architecture: DoubleConv Block Implementation

Design Philosophy and Historical Context To provide an alternative feature extraction approach, we also explored 3D CNN architecture design based on DoubleConv modules. The DoubleConv design concept originally stemmed from the fundamental building blocks of U-Net architecture Ronneberger et al. [2015b], with its core

philosophy centered on enhancing feature representation capabilities through consecutive convolution operations. This architectural pattern has been widely adopted in medical image segmentation tasks, where the sequential convolution design has proven effective for capturing hierarchical features in biological imaging [Jha et al. \[2020\]](#).

Technical Implementation and Architecture Details Unlike residual blocks that employ skip connections, DoubleConv modules adopt a sequential feature transformation strategy, where each module contains two consecutive 3D convolution layers, each followed by ReLU activation and GroupNorm3D normalization. The canonical implementation of DoubleConv follows the pattern (convolution => [BN] => ReLU) * 2, as established in the original U-Net framework and subsequently refined in various medical imaging applications [Jha et al. \[2020\]](#). Recent implementations have incorporated GroupNorm3D instead of BatchNorm for improved performance in 3D medical imaging scenarios, particularly when dealing with small batch sizes common in volumetric data processing [Zhang et al. \[2022\]](#).

Isotropic Convolution Strategy and Design Rationale The motivation behind this design lies in progressively refining cellular features through sequential feature transformations, rather than relying on residual connections to preserve original information. Unlike the anisotropic approach adopted in the residual block variant, our DoubleConv implementation employs isotropic convolution kernels, utilizing $3 \times 3 \times 3$ convolutions consistently across all scales. This design choice simplifies the architecture while maintaining the core philosophy of sequential feature learning. Each DoubleConv module follows the pattern of GroupNorm3D normalization, followed by two consecutive $3 \times 3 \times 3$ convolution transformations, and finalized by a ReLU activation. This design provides rich feature representation capabilities while maintaining computational efficiency. The isotropic kernel design ensures uniform feature extraction across all spatial dimensions, which can be particularly beneficial when the input data preprocessing has already addressed anisotropic characteristics or when computational simplicity is prioritized. [Wolny et al. \[2020b\]](#)

Comparative Analysis with Residual Architectures Compared to the skip connection mechanism of residual blocks, DoubleConv adopts a more direct feature learning pathway, enabling the network to focus on learning hierarchical representations of local features. In cell matching tasks, this design is particularly well-suited for capturing subtle morphological variations in cells, as sequential feature transformations can progressively abstract different levels of morphological features. This sequential approach has demonstrated effectiveness in medical image segmentation tasks where fine-grained feature extraction is crucial [Rehman et al. \[2024\]](#).

Feature Compression Configuration and Performance Considerations The 512-dimensional feature compression was chosen based on GAT framework require-

ments, where optimal performance was achieved when node feature dimensions remained below 1024 to ensure efficient attention computation and avoid over-parameterization issues.

Architectural Integration and Feature Learning Mechanisms As illustrated in Figure 3.3, the DoubleConv-based architecture maintains the same autoencoder structure as the residual variant, compressing 32^3 input image blocks into 512-dimensional feature vectors. The fundamental difference lies in the feature transformation mechanism: while residual blocks facilitate gradient flow through skip connections, DoubleConv modules rely on sequential convolution operations to build feature hierarchies. This architectural choice reflects different philosophical approaches to feature learning—residual connections emphasize information preservation and gradient flow, while DoubleConv emphasizes progressive feature abstraction through direct transformations.

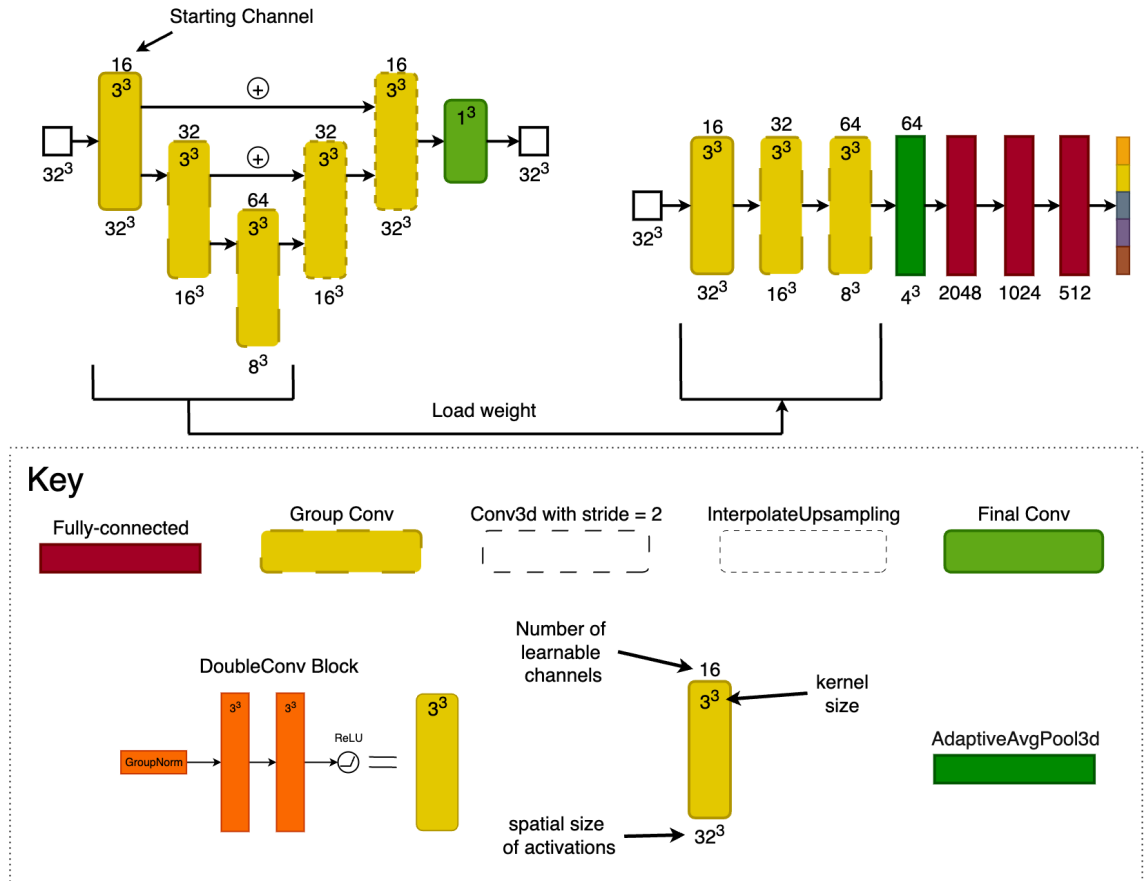


Figure 3.3: Left: 3D CNN Auto-encoder (DoubleConv Block), Right: Encoder + MLP

Advanced Architectural Components

DilatedDenseBlock in the Network Bottleneck A core innovation of the U-Net architecture employed in this research is the introduction of a Dilated Dense Block (DilatedDenseBlock) at the network bottleneck. This design aims to enhance the model’s contextual awareness and multi-scale feature fusion capabilities at the deepest level of feature extraction. This section provides a detailed discourse on its original design principles, its suitability for the domain of 3D medical imaging, and the rationale for its placement within the network bottleneck.

Original Design and Conception The DilatedDenseBlock is not a singular, original concept but rather a hybrid architecture that ingeniously merges two powerful ideas from convolutional neural networks: Dilated Convolution and Dense Connection.

Suitability for 3D Medical Imaging The analysis of 3D medical images, such as for tumor segmentation or organ identification, presents unique challenges, including high variability in the size and shape of target objects and a stringent requirement for precise spatial localization. The design characteristics of the DilatedDenseBlock make it particularly well-suited to address these challenges.

Its key advantage lies in its ability to acquire multi-scale features without sacrificing spatial resolution. In traditional convolutional networks, enlarging the receptive field typically relies on sequential pooling or downsampling operations, which inevitably leads to a loss of spatial information. This is a significant drawback for medical imaging tasks that demand precise boundary segmentation. The DilatedDenseBlock addresses this by using dilated convolutions to capture a wide range of information, from local details to global tissue structures, on feature maps that maintain a higher spatial resolution.

As demonstrated in the work on a Dilated Dense U-Net for infant hippocampus segmentation, the core value of such a structure lies in its capacity to provide multi-scale features while preserving high spatial resolution [Li et al. \[2019\]](#). This allows the model to simultaneously perceive minute pathological details and their surrounding anatomical context, leading to more accurate judgments.

Rationale for Placement in the Network Bottleneck In encoder-decoder architectures like U-Net, the bottleneck layer plays a pivotal role. It is situated at the deepest point of the network, where feature maps have the lowest spatial resolution but the highest level of semantic abstraction. Placing the DilatedDenseBlock at this juncture is justified by several key rationales:

- **Maximizing Receptive Field Efficiency:** Since the bottleneck has the smallest spatial dimensions, it is the optimal location to achieve a global receptive field with minimal computational cost. Enabling the model to have a comprehensive contextual understanding of the entire input image before

beginning feature map reconstruction in the decoder is crucial for improving segmentation accuracy.

- **Enhancing Deep Feature Fusion:** The bottleneck serves as the transition point where the network shifts from feature extraction to segmentation reconstruction. Introducing a DilatedDenseBlock here allows for an intensive, multi-scale fusion and reprocessing of the highest-level abstract features, thereby providing richer and more discriminative feature information for the upsampling path of the decoder.

This design strategy is echoed in multiple studies. For instance, a model for carotid plaque segmentation explicitly incorporates six dilated convolution layers in its bottleneck section, in contrast to the three standard convolutional layers in a traditional U-Net [Lekadir et al. \[2021\]](#). Furthermore, the data flow described by [Li et al. \[2019\]](#) validates this placement, where feature maps from the encoder are processed by the dilated dense network before their outputs are concatenated with the corresponding feature maps in the expansive path.

Dilated Convolution As proposed by [Yu and Koltun \[2016\]](#), aims to exponentially expand the receptive field of convolutional kernels without increasing computational parameters or reducing spatial resolution. By strategically inserting "holes" or gaps between the weights of a kernel, a single convolutional layer can capture a wider range of contextual information, which is critical for understanding the global structure of an image.

Dense Connection Originating from the DenseNet architecture by [Huang et al. \[2017\]](#), is based on the principle of connecting each layer to every other layer in a feed-forward fashion within a block. This design significantly promotes feature reuse, enhances feature propagation, and alleviates the vanishing gradient problem in deep networks.

The DilatedDenseBlock is the product of combining these two concepts. It leverages a densely connected architecture to improve information flow and feature reuse, while employing dilated convolutions with varying rates in each layer. This enables the block to efficiently aggregate multi-scale features, thereby achieving a richer and more comprehensive understanding of the input data.

In summary, employing the DilatedDenseBlock as the U-Net bottleneck is a highly efficient and academically validated design. It fully leverages the characteristics of the bottleneck layer to achieve global contextual awareness and deep fusion of multi-scale features at a minimal cost, thereby significantly enhancing the model's performance on complex medical image segmentation tasks.

3.2.4 Rotation-Invariant: CubeNet

Group Equivariant Theory Application

Limitations of Traditional CNN Approaches Traditional CNNs relied solely on data augmentation to approximate rotation invariance through statistical learning, which required extensive training data and could not guarantee robust performance on unseen orientations. In contrast, Worrall and Brostow [2018] offered a principled approach to achieving 3D rotation equivariance by incorporating group-theoretic principles directly into the network architecture.

T4 Rotational Group and Mathematical Framework CubeNet employed group convolutions based on the rotational tetrahedral group (T4), which consisted of twelve specific rotations from the complete set of 24 cubic rotations. This subgroup included the identity rotation and eleven additional rotations that corresponded to all even rotations (rotations that could be composed of an even number of 90° rotations). The T4 group encompassed rotations that preserved the tetrahedral symmetry within the cube, providing a more comprehensive coverage of 3D orientations compared to smaller subgroups while maintaining computational tractability. The key advantage of this approach lay in its **linear equivariance property**: when the input underwent a rotation from the group, the feature representations were transformed in a predictable, linear manner. This was mathematically expressed as:

$$\Phi(gx) = g\Phi(x)$$

where Φ represented the feature extraction function, g was a group element (rotation), and x was the input.

Weight-Sharing Mechanism and Computational Advantages The weight-sharing mechanism in group convolutions ensured that the same filter weights were applied to rotated versions of the input, effectively learning rotation-invariant features without requiring explicit data augmentation. This was particularly beneficial for cellular imaging applications where cells could appear in arbitrary orientations, and traditional CNNs would need extensive rotational data augmentation to achieve similar robustness. The T4 group provided a good balance between computational efficiency and rotational coverage, as it captured the essential 3D rotational symmetries while using only 12 orientations instead of the full 24.

Two-Stage Training Architecture Implementation As illustrated in Figure 3.5, our rotation-invariant architecture followed a two-stage training approach. The left side showed the Group Convolutional 3D (Gconv3d) Autoencoder used for pre-training the encoder weights, where the group dimension of 12 corresponded to the T4 rotational group. This pre-training stage enabled the encoder to learn robust, rotation-equivariant feature representations from cellular image data across

all 12 tetrahedral rotations. The right side demonstrated the subsequent architecture where the pre-trained encoder was combined with an MLP to compress cellular features into 512-dimensional representations optimized for matching tasks.

Integration Benefits and Performance Advantages The integration of CubeNet’s T4 group convolution principles into our autoencoder architecture provided several advantages: first, it reduced the dependency on extensive rotational data augmentation during training; second, it ensured consistent feature extraction regardless of cellular orientation across the 12 tetrahedral rotations; third, it maintained computational efficiency compared to exhaustive rotation averaging at test time while providing more comprehensive rotational coverage than smaller groups like V4. This rotation-invariant feature extraction formed the foundation for robust cellular matching across different imaging conditions and orientations.

CubeNet Implementation Challenges

The implementation of CubeNet’s group equivariant convolutions introduced several technical obstacles in our cellular matching framework. Specifically, we encountered issues related to group dimension matching, computational resource consumption, and the integration of feature-compression layers.

Group Dimension Matching Difficulties The T4 rotational group with 12 elements imposed substantial complexity on tensor dimension alignment. Careful management of feature tensor reshaping between group convolution layers was required, and the subsequent interaction with NdLinear compression layers often led to dimensionality mismatches.

Computational Resource Limitations Group convolutions increased memory demands by a factor of 12, corresponding to the size of the T4 rotational group, significantly prolonging training times compared to standard 3D CNNs. On an NVIDIA A100 GPU with 40GB VRAM, the batch size was limited to 32 samples, due to the increased feature-map storage requirements arising from the additional group dimension. While these constraints were manageable during autoencoder training, applying a multilayer perceptron (MLP) immediately after the encoder posed further VRAM limitations. Specifically, computing features for 558 cells simultaneously exceeded the available GPU memory on our cluster, making such processing infeasible.

Feature Compression Integration Challenges NdLinear compression layers were not originally designed for group equivariant features. Standard pooling operations across group dimensions led to noticeable information loss, diminishing the theoretical benefits of equivariant convolutions.

Alternative Solutions and Workarounds

To address these implementation challenges, several alternative solutions and workarounds were developed and tested. A modified group convolution architecture was implemented that incorporated group-aware pooling mechanisms to reduce computational overhead. Custom tensor reshaping functions were developed to enable seamless integration with NdLinear compression, while gradient accumulation techniques were applied to handle the constraints imposed by smaller batch sizes.

Computational Optimization Strategies Computational optimization strategies were employed to mitigate resource limitations. Mixed precision training using FP16 was implemented to reduce the memory footprint, and checkpoint gradient computation was utilized to trade memory requirements for increased computation time. Additionally, progressive training with increasing group complexity was adopted to gradually build model capacity while managing computational demands.

Architecture Adaptations Architecture adaptations were necessary to accommodate the increased complexity. The network depth was reduced to compensate for the increased group dimension complexity, and early group dimension reduction was implemented in the encoder pathway. The bottleneck architecture was modified specifically to handle group equivariant features more effectively.

Implementation Outcomes and Model Integration These optimization strategies enabled the partial implementation of CubeNet within our computational constraints, allowing for preliminary evaluation of group equivariant approaches in cellular matching tasks. The modified architecture was integrated into our overall framework as shown in Figure 3.1 (dashed box), where it served as an alternative rotation-invariant module alongside traditional data augmentation methods. While computational limitations prevented full-scale deployment, the implemented solutions provided valuable insights into the practical requirements for group equivariant neural networks in biological image analysis.

The resulting implementation served as a proof-of-concept for applying geometric deep learning principles to cellular matching, establishing a foundation for future investigations with enhanced computational resources. These technical developments contribute to the broader understanding of implementing group equivariant architectures in resource-constrained research environments.

3.3 Feature Compression and Generation

In our framework, high-dimensional features extracted by the 3D CNN encoder must be compressed into a lower-dimensional representation suitable for downstream tasks. This section details the two primary strategies we evaluated for this purpose:

a conventional approach using a Multi-Layer Perceptron (MLP) that relies on feature flattening, and an alternative method using `NdLinear` layers that preserves the inherent tensor structure.

3.3.1 Feature Compression Strategy Implementation

MLP-based Compression Pipeline: This traditional compression pathway, based on a Multi-Layer Perceptron (MLP), involves flattening the high-dimensional feature tensor into a vector before dimensionality reduction. The process begins with a feature tensor of shape $(64, 8, 8, 8)$ from our standard 3D CNN encoder, or $(64, 12, 8, 8, 8)$ from the CubeNet encoder. In these tensors, 64 represents the feature channels, 12 denotes the group dimension for CubeNet, and $(8, 8, 8)$ are the compressed spatial dimensions.

The MLP-based pathway proceeds through the following sequential operations:

1. **Group-wise Pooling (for CubeNet features):** For features originating from CubeNet, a pooling operation (e.g., `max` or `mean`) is optionally applied across the group dimension. This step can standardize the tensor shape to $(64, 8, 8, 8)$ for subsequent processing.
2. **Spatial Pooling:** A 3D adaptive average pooling layer (`AdaptiveAvgPool3d`) is applied to reduce the spatial dimensions, transforming the tensor into a shape of $(64, 4, 4, 4)$.
3. **Flattening Operation:** The resulting 3D feature map is flattened into a one-dimensional vector. For a standard input tensor that has been pooled, this results in a vector of length $64 \times 4 \times 4 \times 4 = 4,096$. If group-wise pooling is omitted for CubeNet features, the vector length becomes $64 \times 12 \times 4 \times 4 \times 4 = 49,152$.
4. **MLP Processing:** This high-dimensional vector is subsequently processed by an MLP, which serves as a projection head to produce the final compact feature representation. The MLP architecture is defined as follows:
 - **Input Layer:** 4,096 dimensions (or 49,152 for unabridged CubeNet features).
 - **Hidden Layer 1:** 2,048 dimensions (or 8,192), followed by a ReLU activation.
 - **Hidden Layer 2:** 1,024 dimensions (or 2,048), followed by a ReLU activation.
 - **Output Layer:** 512 dimensions, yielding the final feature representation.

NdLinear-based Compression Pipeline: In contrast to the MLP-based approach, the `NdLinear` compression pipeline is designed to preserve the inherent N-dimensional structure of the feature tensor by applying dimension-wise transformations, thereby

avoiding the information loss associated with premature flattening [Reneau et al. \[2025\]](#).

1. **Direct Tensor Processing:** The input feature tensor, with shape (64, 8, 8, 8) or (64, 12, 8, 8, 8), is directly fed into a projection head composed of multiple NdLinear layers without any initial flattening.
2. **Progressive Compression:** A sequence of 2 to 4 NdLinear layers is applied to progressively compress the tensor along its dimensions. An example reduction strategy is as follows:
 - **Layer 1:** (64, 12, 8, 8, 8) $\rightarrow$ (64, 12, 4, 4, 4)
 - **Layer 2:** (64, 12, 4, 4, 4) $\rightarrow$ (16, 2, 4, 4, 4)
 - **Layer 3:** (16, 2, 4, 4, 4) $\rightarrow$ (4, 2, 4, 4, 4)
3. **Delayed Flattening:** The flattening operation is deferred until the final stage. Only after sufficient dimensional reduction is the structured, low-dimensional tensor (e.g., (4, 2, 4, 4, 4)) unrolled into a 512-dimensional vector. This strategy maximally preserves structural information throughout the compression process and provides the final representation for downstream matching tasks[1].

3.3.2 Loss Function Selection and Application Strategy

Loss Selection for Reconstruction Tasks: For the autoencoder’s reconstruction task, the choice of loss function is critical for learning high-quality feature representations. Based on theoretical analysis, L1 loss (Mean Absolute Error) is favored over Mean Squared Error (MSE). Its linear penalty structure is less sensitive to outliers and demonstrates superior performance in preserving the sharp edges and fine details of cellular structures, which can be blurred by the quadratic penalty of MSE. The mathematical formulations are as follows:

- **L1 Loss (MAE):** $\mathcal{L}_{L1} = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|$
- **MSE Loss:** $\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$

where x_i is the original input and $\hat{x}_i$ is the reconstructed output.

Loss Selection for Cellular Matching Tasks: For the downstream cellular matching task, a distance metric is required to compute the cost matrix between feature representations of cells from two different specimens. To establish a robust baseline and maintain computational efficiency, we employ the Mean Squared Error (MSE) to quantify the distance between two feature vectors, h_i and h_j , from a pair of *C. elegans* samples. The distance is calculated as:

$$d_{MSE}(h_i, h_j) = \frac{1}{D} \sum_{d=1}^D (h_i^d - h_j^d)^2$$

where D is the dimensionality of the feature vectors.

3.4 Training and Optimization Framework

3.4.1 Training Strategy

Permutation-based Training for Positional Invariance To ensure that the neural network relied solely on cellular features for matching decisions rather than positional biases, we implemented a permutation-based training strategy. During training, we randomly permuted the order of cell pairs presented to the network while maintaining their ground truth correspondence labels.

Specifically, for each training batch containing N cell pairs $(c_i^{(1)}, c_i^{(2)})$ where $i \in \{1, 2, \dots, N\}$, we applied random permutations π_1 and π_2 to the first and second sets of cells respectively:

$$\{(c_{\pi_1(i)}^{(1)}, c_{\pi_2(j)}^{(2)}) | i, j \in \{1, 2, \dots, N\}\}$$

The network had to then learn to identify correct correspondences based purely on the extracted cellular features, regardless of the presentation order. This permutation strategy significantly improved the generalization capability of the matching system and ensured robust performance across different cell arrangements and imaging conditions.

Architectural Flexibility The integration of DoubleConv modules with the permutation-based training strategy ensures that the network learns robust cellular feature representations regardless of the specific architectural choice. Both architectural variants—residual (Figure 3.4) and DoubleConv—demonstrate (Figure 3.3) the flexibility of the proposed framework in accommodating different feature extraction paradigms while maintaining consistent performance in cell matching tasks.

Learning Matching-Oriented Features with LAP As shown in Figure 3.4 and Figure 3.3, the right-side architecture adopted feature representations after the encoder and combined Linear Assignment Problem (LAP) to learn matching feature extraction. The goal of this design was to compress complex 3D cell features into 512-dimensional compact representations while maintaining sufficient discriminability to support subsequent cell matching tasks. The introduction of LAP enabled the network to learn feature representations more suitable for matching tasks. Compared to traditional reconstruction loss, this supervision signal more directly targeted the core tasks of cell identification and correspondence establishment. Figure 3.4 provides a comprehensive overview of the overall architecture, illustrating how residual-based feature extraction approaches integrate with the subsequent matching pipeline.

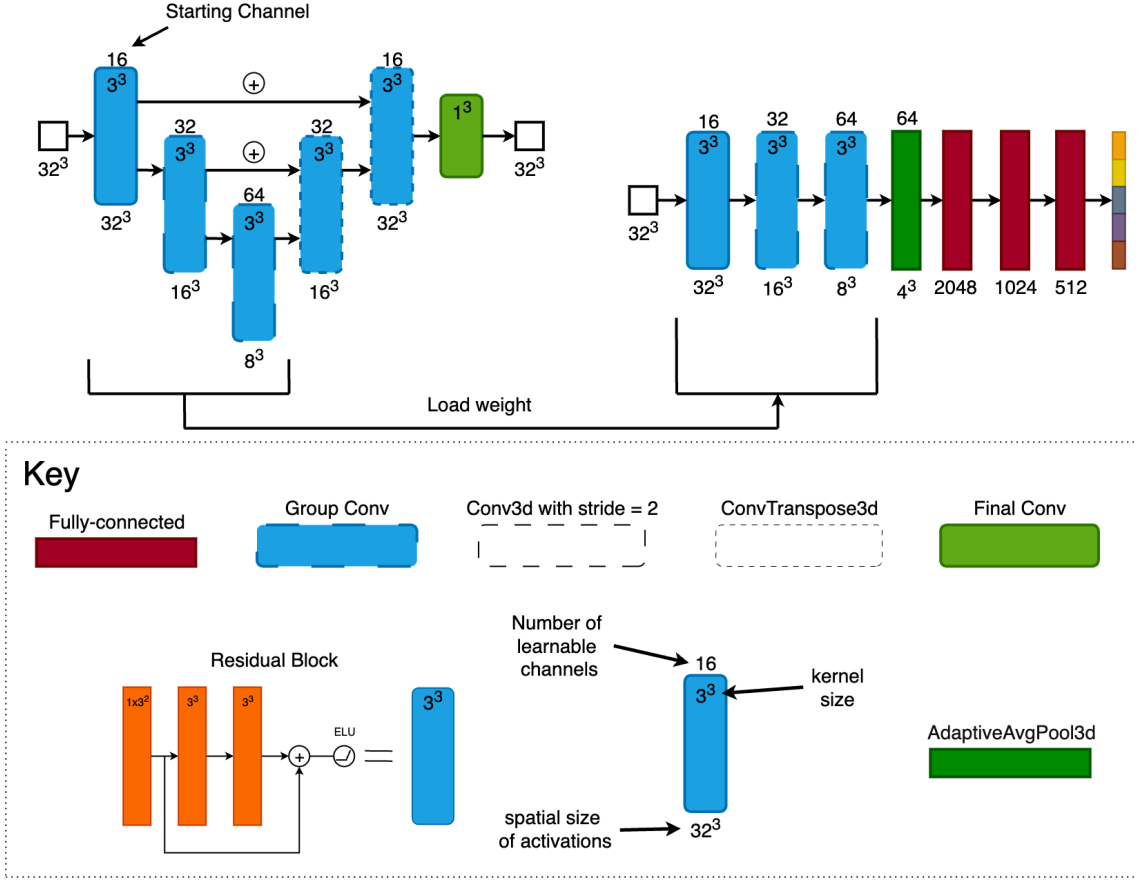


Figure 3.4: Left: 3D CNN Auto-encoder (Residual Block), Right: Encoder + MLP

3.4.2 Loss Function Implementation

Loss Function Implementation: The training employs a multi-component loss function balancing reconstruction quality and matching accuracy:

$$\mathcal{L}_{total} = \mathcal{L}_{matching}$$

where:

- $\mathcal{L}_{matching}$: Differentiable Hungarian loss for assignment optimization

Differentiable Hungarian Algorithm Implementation:

1. **Forward Pass Hungarian Solver:** The Hungarian algorithm solves the linear assignment problem:

$$\mathbf{P}^* = \arg \min_{\mathbf{P} \in \Pi_n} \langle \mathbf{C}, \mathbf{P} \rangle = \arg \min_{\mathbf{P} \in \Pi_n} \sum_{i,j} C_{ij} P_{ij}$$

where Π_n represents the set of $n \times n$ permutation matrices.

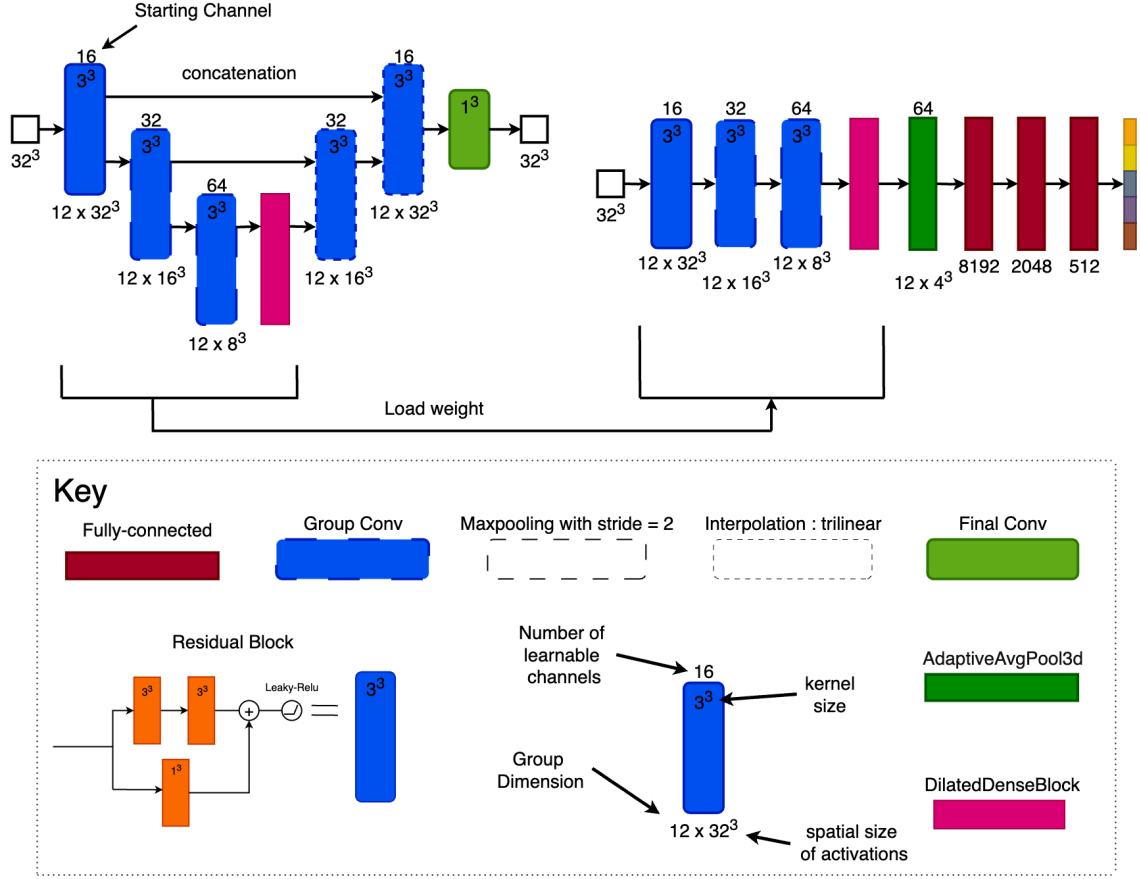


Figure 3.5: Left: Gconv3d Auto-encoder, Right: Encoder + MLP

2. **Gradient Estimation via Perturbation:** The backward pass approximates gradients using perturbation technique:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{C}} \approx - \frac{\mathbf{P}^*(\mathbf{C}) - \mathbf{P}^*(\mathbf{C} + \lambda \nabla_{\mathbf{P}} \mathcal{L})}{\lambda}$$

3. **Lambda Parameter Configuration:**

- **Initial Value:** $\lambda = 1.0$ for stable gradient estimation
- **Range Testing:** Experimented with $\lambda \in \{0.1, 0.5, 1.0, 2.0, 5.0, 50, 100, 200\}$
- **Optimal Selection:** $\lambda = 1.0$ provided best trade-off between gradient accuracy and computational stability

4. **Implementation Details:**

- Hungarian solver: SciPy's `linear_sum_assignment` for forward pass
- Gradient computation: Custom PyTorch autograd function
- Batch processing: Parallel Hungarian solving for multiple graph pairs

3.5 Graph-based Contextual Processing

3.5.1 Graph Attention Networks

Building upon the theoretical foundations established in the background section, we implemented Graph Attention Networks to process the integrated feature representations from the previous stages. The choice of GAT over traditional Graph Convolutional Networks was motivated by the dynamic nature of graph construction in our framework. Since each graph was constructed independently for individual organisms, the same cell type across different samples would not maintain identical edge connectivity patterns. This variability in graph topology necessitated the use of GAT’s adaptive attention mechanism, which could dynamically adjust to different neighborhood structures, unlike GCN’s fixed aggregation weights that assumed consistent graph topology. [Veličković et al. \[2017\]](#)

The graph construction process employed a radius-based connectivity approach optimized for cellular matching applications. Based on empirical analysis presented in Figure 4.1, we selected an optimal radius parameter that balanced graph connectivity with computational tractability. The complete graph construction methodology and parameter selection rationale are detailed in the Graph Connectivity Strategy section below.

Node Feature Composition Each node $v_i \in V$ represented an individual cell with features combining multiple information sources:

$$h_i = [f_i^{deep}, f_i^{struct}, f_i^{geo}] \in \mathbb{R}^{512+8}$$

where $f_i^{deep} \in \mathbb{R}^{512}$ represented the learned cellular features from either the standard 3D CNN autoencoder or the rotation-invariant group convolutional autoencoder, $f_i^{struct} \in \mathbb{R}^3$ contained structural features (node degree, clustering coefficient, PageRank), and $f_i^{geo} \in \mathbb{R}^5$ included geometric descriptors (distance to centroid, relative positions in x, y, z dimensions).

Graph Connectivity Strategy We established graph connectivity using a radius-based approach with radius $r = 45$ units, ensuring that cells within biologically meaningful distances were connected:

$$E = \{(i, j) \mid \|pos_i - pos_j\|_2 \leq r\}$$

This parameter was selected based on comprehensive empirical analysis presented in Figure 4.1. Radius values of 60 resulted in excessive edge density, leading to computational inefficiency and diminishing returns in classification accuracy. Conversely, radius values of 30 produced too many nodes with minimal connectivity (only 3 edges), creating isolated subgraphs that hindered effective message passing.

The chosen radius of 45 represented an optimal balance between graph connectivity and computational tractability, achieving substantial connectivity improvements while maintaining manageable computational overhead.

To ensure graph connectivity and prevent isolated nodes, we implemented a minimum degree constraint where nodes with fewer than 3 connections were supplemented with k-nearest neighbor connections. This strategy guaranteed that all cellular nodes could participate meaningfully in the graph attention mechanism.

Feature Standardization All feature components underwent standardization using StandardScaler to ensure balanced contribution across different feature types:

$$f_i^{normalized} = \frac{f_i - \mu_f}{\sigma_f}$$

where μ_f and σ_f were the mean and standard deviation of each feature dimension across the dataset. This normalization step was crucial for preventing any single feature type from dominating the attention computation in subsequent GAT layers.

3.5.2 Multi-Head GAT Architecture

Building upon the node feature representations established in the previous subsection, we designed a sophisticated two-layer GAT architecture. This network leverages multi-head attention mechanisms to capture diverse cellular relationships and spatial dependencies, while being carefully engineered to balance representational capacity with computational efficiency. Veličković et al. [2017]

Network Structure The core of our GAT model is composed of two sequential layers that progressively refine the node features. The specific architecture is as follows:

- **Layer 1:** A multi-head attention layer with 8 heads, which transforms the 520-dimensional input features into a 256-dimensional hidden representation for each head.
- **Layer 2:** A final single-head attention layer that aggregates the multi-head outputs from the previous layer into a conclusive 128-dimensional feature vector for each node.

Stabilization and Regularization To ensure robust training dynamics and enhance generalization, we incorporated several key design elements throughout the architecture:

- **Residual Connections:** Implemented between the input and output of the GAT module to facilitate gradient flow, preventing the vanishing gradient problem in deeper graph structures He et al. [2016b].

- **Layer Normalization:** Applied after each GAT layer to stabilize the learning process and improve convergence properties Ba et al. [2016].
- **ELU Activation Functions:** Chosen for their ability to handle negative values and provide smoother activation surfaces, which can improve gradient flow compared to standard ReLU functions Clevert et al. [2015].
- **Dropout Regularization:** Applied between layers with a configurable rate to prevent overfitting by reducing co-adaptation between neurons Srivastava et al. [2014a].

3.5.3 Training Objective and Optimization

The training methodology seamlessly integrates the architectural components described above through a sophisticated end-to-end optimization strategy. The GAT network was trained using a differentiable Hungarian loss that directly optimized for matching accuracy through a combination of assignment optimization and matching evaluation:

$$\mathcal{L} = \mathcal{L}_{Hamming}(\mathbf{P}_{predicted}, \mathbf{P}_{ideal})$$

where the loss computation involved several interconnected components that work together to achieve optimal cellular matching performance.

Cost Matrix Computation The cost matrix $\mathbf{C}$ between learned representations H_1 and H_2 was computed using configurable distance metrics:

$$\mathbf{C}_{ij} = d(h_i^{(1)}, h_j^{(2)})$$

where $d(\cdot, \cdot)$ employed MSE distance to quantify feature similarity between cellular representations from different organisms.

Differentiable Hungarian Assignment The optimal assignment was computed using a differentiable approximation of the Hungarian algorithm through perturbation-based gradient estimation:

$$\mathbf{P}_{predicted} = \text{Hungarian}(\mathbf{C})$$

The backward pass employed a perturbation technique where the cost matrix was modified by $\lambda \nabla \mathcal{L} / \nabla \mathbf{P}$ to compute gradients:

$$\mathbf{C}' = \mathbf{C} + \lambda \frac{\partial \mathcal{L}}{\partial \mathbf{P}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{C}} = - \frac{\mathbf{P}_{forward} - \mathbf{P}_{backward}}{\lambda}$$

where $\mathbf{P}_{forward}$ and $\mathbf{P}_{backward}$ were the assignments computed on the original and perturbed cost matrices respectively.

Hamming Loss Evaluation The final loss was computed using Hamming distance between the predicted assignment matrix and the ideal ground truth assignment:

$$\mathcal{L}_{Hamming} = \frac{1}{N^2} \sum_{i,j} [\mathbf{P}_{predicted}(i,j) \cdot (1 - \mathbf{P}_{ideal}(i,j)) + (1 - \mathbf{P}_{predicted}(i,j)) \cdot \mathbf{P}_{ideal}(i,j)]$$

where $\mathbf{P}_{ideal}$ was constructed from the ground truth permutation mappings between the two graphs.

Dynamic Graph Pairing Strategy During training, we employed a dynamic pairing strategy where each graph was randomly paired with another graph from the dataset, ensuring diverse training combinations and preventing overfitting to specific graph pairs. This strategy enhanced the model’s generalization capability across different cellular configurations and imaging conditions.

Permutation Invariance To ensure the model learned genuine cellular correspondences rather than positional biases, we applied random permutations to node ordering during graph construction while maintaining ground truth correspondence information through inverse permutation mappings. This approach guaranteed that the learned representations captured intrinsic cellular features rather than arbitrary ordering dependencies.

3.5.4 Integration Pipeline

The final integration pipeline synthesizes all the components described in the previous subsections to deliver robust cellular matching capabilities. The learned node representations from GAT served as the final feature vectors for cellular matching, incorporating both individual cellular characteristics and contextual spatial information.

The matching process employed the differentiable Hungarian algorithm to solve the linear assignment problem:

$$C_{ij} = d(h'_i, h'_j)$$

where $d(\cdot, \cdot)$ represented either MSE distance or cosine distance between GAT-processed node features h'_i and h'_j . This final matching stage leveraged the enriched representations produced by the graph attention mechanism to make more informed correspondence decisions.

The training process incorporated several optimization strategies that ensured robust convergence and optimal performance:

- **Adam optimizer** with learning rate 0.0005 for stable gradient-based optimization

- **Early stopping** with patience of 100 epochs monitoring validation loss to prevent overfitting
- **Model checkpointing** saving the top 3 models based on validation performance for robust model selection
- **Optional lambda scheduling** for dynamic loss weighting during training to balance different optimization objectives

This comprehensive approach ensured that the final matching decisions considered both individual cellular features learned through deep 3D CNN encoders and spatial contextual information captured through the graph attention mechanism, leading to more accurate and biologically meaningful cell correspondences. The integration of all these components creates a robust framework capable of handling the complex spatial relationships and morphological variations inherent in cellular matching tasks.

3.6 Implementation Details

Hardware Configuration: Except for the cells feature learning experiments with MLP, which were conducted on a computing system equipped with an NVIDIA A100 GPU (40GB VRAM) and AMD EPYC 7513 CPU, all other experiments were performed on a system with an NVIDIA GTX 1080 Ti GPU (Pascal, 11GB memory).

Parameter Settings and Software Implementation: Detailed parameter configurations, software dependencies, and implementation code are available in our GitHub repository: [MA-FeatureLearning](#)

4 Experiments

4.1 Feature Extraction Layer Comparison

Our feature extraction strategy was inspired by Discrete Cycle-Consistency Based Unsupervised Deep Graph Matching [Tourani et al. \[2024\]](#), which utilizes the outputs of the $relu4_2$ and $relu5_1$ operations of the VGG16 network as matching features. To investigate the impact of different feature extraction depths on matching performance, we compared the use of features from the encoder’s last layer versus the last two layers.

As shown in Table 4.1, configurations using features from the last two layers generally outperformed single-layer feature extraction in most cases. Particularly, under the Starting Channel 16, 3-layer network configuration with MSE loss function, the last two layers feature extraction achieved the highest accuracy of **31.70%**, representing a significant improvement compared to the corresponding single-layer configuration (25.20%). This result demonstrates that multi-layer feature fusion can provide richer semantic information, contributing to improved cell matching accuracy.

However, as the network depth increased to 4 layers, the performance of last two layers feature extraction declined, which may be attributed to overfitting caused by over-parameterization. While experimental results support the adoption of multi-layer feature fusion strategies, **considering that an ideal feature compression method has not yet been identified and constrained by computational resource limitations**, subsequent experiments in this study will focus on single-layer feature extraction optimization. Although multi-layer feature fusion theoretically

Table 4.1: Comparison of linear assignment matching accuracy (pre-trained encoder with L1 and MSE loss) for two *C. elegans*, using features extracted from the last one or last two encoder layers of autoencoders trained with different starting channel numbers and overall layer depths.

		Starting Channel: 8		Starting Channel: 16	
		L1	MSE	L1	MSE
Last one	Overall Layers				
	3	21.68	25.95	21.29	25.20
Last two	4	12.49	24.59	25.30	28.48
	3	26.42	27.80	24.86	31.70
	4	20.09	18.33	19.91	24.98

possesses stronger representational capabilities, the increased computational complexity and memory requirements make large-scale experiments impractical. Therefore, until effective feature compression techniques are discovered, we will pursue single-layer feature extraction as our primary research direction and explore other improvement strategies based on this foundation.

4.2 Geometric and Structural Feature Baseline Analysis

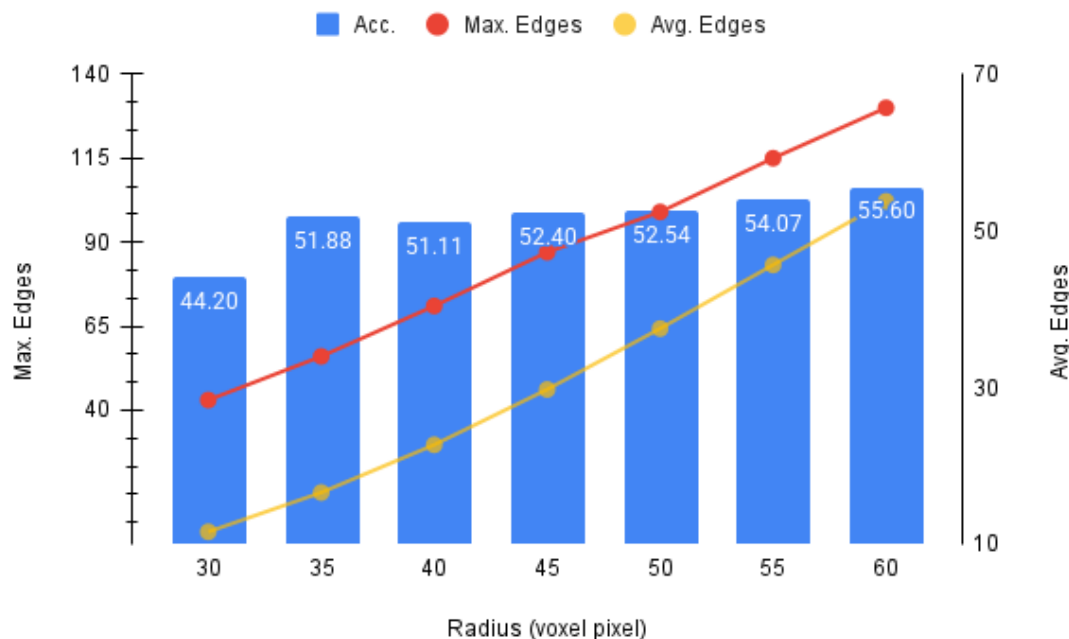


Figure 4.1: Graph statistics and classification accuracy ($Acc.$) for *C. elegans* cell graphs constructed with varying connection radius (voxel pixel). Each graph is built by connecting cells within a given radius; if a cell has fewer than three neighbors, k-nearest neighbors (kNN) are used to ensure a minimum degree of three. *Max. Edges* (red circles) and *Avg. Edges* (orange circles) represent the maximum and average number of neighbors (edges) per cell, respectively, plotted against the right y-axis. *Acc.* (blue bars) reports the classification accuracy using only the geometric and structural features derived from the graph, with values displayed on top of each bar and plotted against the right y-axis.

To establish a comparison baseline, we first evaluated the Linear Assignment Problem (LAP) performance using only geometric and structural features. Figure

4.1 demonstrates the relationship between statistical properties of *C. elegans* cell graphs constructed with varying connection radii and classification accuracy.

As observed from the figure, graph connectivity significantly enhanced as the connection radius increased from 30 to 60 pixels. The maximum number of edges (Max. Edges) increased from approximately 30 to over 65, while the average number of edges (Avg. Edges) exhibited a similar upward trend. Correspondingly, the classification accuracy using purely geometric and structural features steadily improved from 44.20% to 55.60%.

However, careful analysis of the accuracy trend reveals a critical consideration for optimal radius selection. While connectivity continues to increase beyond radius 45, the marginal improvement in classification accuracy becomes negligible. Specifically, increasing the radius from 45 to 60 pixels results in substantial computational overhead—with maximum edges increasing from approximately 45 to over 65—while accuracy improvement plateaus at only 3.20% (from 52.40% to 55.60%).

Therefore, *radius* = 45 was selected as the optimal compromise between computational efficiency and classification performance. This choice represents the point of diminishing returns where further increases in graph connectivity yield minimal accuracy gains while significantly increasing computational complexity. This trade-off is particularly important for subsequent deep learning stages where computational resources must be allocated efficiently across the entire pipeline.

This result indicates that richer graph connectivity can provide more comprehensive structural information for cell matching. As the connection radius increases, each cell node can perceive more neighboring nodes, thereby obtaining more complete local structural descriptions. This provides an important reference baseline for subsequent deep learning feature extraction.

4.3 Autoencoder Architecture Design and GNN Integration Effects

4.3.1 Network Architecture Parameter Optimization

We systematically evaluated the impact of three key architectural parameters on model performance: starting channel number (8 vs 16), loss function type (L1 vs MSE), and network depth (3 vs 4 layers).

Starting Channel Number Impact: From Tables 4.2 and 4.4, it can be observed that larger starting channel numbers (16) generally provide better feature representation capabilities. Under the Residual modules configuration, Starting Channel 16 outperformed Starting Channel 8 in most cases, indicating that larger feature dimensions help capture subtle differences between cells.

Loss Function Selection: The relationship between L1 and MSE loss functions reveals a nuanced pattern that depends on both architectural choice and network

configuration. Overall analysis indicates that **L1 loss demonstrates superior performance in the most critical configurations**, particularly achieving the highest overall accuracy of 74.09% with Residual modules (Starting Channel 16, 3-layer with GNN integration), compared to MSE’s 72.76% in the same configuration.

In DoubleConv ConvBlocks architectures, L1 loss consistently outperformed MSE across both GNN and non-GNN configurations. Most notably, L1 achieved 67.81% accuracy versus MSE’s 63.98% in the optimal DoubleConv configuration (Starting Channel 16, 3-layer with GNN). This pattern extends to the baseline non-GNN configurations where L1 reached 14.53% compared to MSE’s 11.38%.

For Residual modules, the performance relationship is more complex but ultimately favors L1 in the most important scenarios. While MSE showed advantages in some Starting Channel 8 configurations (particularly 72.58% vs 64.52% in 3-layer GNN), **L1 achieved the global optimum** in the Starting Channel 16, 3-layer configuration with 74.09% accuracy.

This suggests that **L1 loss is better suited for cellular matching tasks**, likely due to its linear penalty structure that preserves fine cellular details without over-penalizing moderate feature differences. The superior performance of L1 in achieving the highest overall accuracy demonstrates its effectiveness in learning discriminative cellular representations for matching applications.

Network Depth: 3-layer network configurations consistently out-performed their 4-layer counterparts. This result aligns with findings in the 3D-autoencoder literature showing that **once the encoder–decoder stack grows beyond roughly three convolutional blocks, high-frequency structures are progressively smoothed out and reconstructed volumes lose fine anatomical detail.**

Empirical evidence includes:

- **Over-smoothing in deep 3D autoencoders** – Özgün Çiçek et al. [2016] extended a 3-block 3D U-Net autoencoder to five blocks; although global similarity scores improved, thin trabecular structures in micro-CT scans became visibly blurred due to cumulative interpolation and vanishing-gradient effects that suppress high-frequency features.
- **Detail loss beyond three layers** – In brain-MRI super-resolution experiments, Pham et al. [2022] compared 2-, 3- and 4-layer 3D autoencoders: the 4-layer model showed a 6 % drop in SSIM for cortical ribbon regions and produced noticeably softer edges, whereas the 3-layer variant preserved gyral contours best.
- **Depth-induced blurring in volumetric reconstruction** – Hao et al. [2021] demonstrated that adding more than three residual down-sampling stages to a voxelwise autoencoder increased PSNR but reduced the surface-based Dice score from 0.84 to 0.76 for small vessels, concluding that “deeper networks trade local detail for global consistency” in 3-D volume reconstructions.

These findings support our choice of a shallower (3-layer) architecture: deeper stacks risk over-regularisation and gradient attenuation, which together erode the sub-voxel information critical for precise cell-level matching in *C. elegans*.

4.3.2 GNN Integration Performance Enhancement

Graph Neural Network (GNN) integration brought significant performance improvements. By comparing configurations without GNN (Tables 4.2, 4.4) with GNN configurations (Tables 4.3, 4.5), we found:

- **Residual modules:** After GNN integration, accuracy improved from a maximum of 18.06% to 74.09%, representing an enhancement of 56.03%. This optimal performance was achieved using the L1 loss function with starting channel 16 configuration, demonstrating the effectiveness of residual connections in preserving gradient flow for cellular feature learning.
- **DoubleConv ConvBlocks:** After GNN integration, accuracy improved from a maximum of 14.53% to 67.81%, representing an enhancement of 53.28%

This substantial performance improvement demonstrates the critical role of Graph Neural Networks in handling cell matching tasks. GNNs can effectively aggregate information from neighboring nodes, learning more expressive graph-level feature representations.

Architecture Comparison: Residual modules architecture overall outperformed DoubleConv ConvBlocks, which may be due to residual connections’ ability to alleviate gradient vanishing problems and promote better feature learning.

Table 4.2: Linear assignment matching accuracy using **Residual modules** without GNN. Results show 512-dimensional MLP features obtained from multi-layer perceptron with L1 and MSE loss across different starting channel sizes and network depths.

(a) Starting Channel 8 results			(b) Starting Channel 16 results		
Layers	L1	MSE	Layers	L1	MSE
3	17.74	15.70	3	16.47	18.06
4	11.51	13.14	4	13.48	17.46

4.4 Comparison with Existing Methods

Table 4.6 presents the comparison results between our method and existing state-of-the-art approaches. Our method achieved an accuracy of 74.09%, while Active

Table 4.3: Linear assignment matching accuracy using **Residual modules** with GNN integration. Features combine 512 MLP features with 8 geometric and structural features (total 520) processed by Graph Attention Network across different starting channel sizes and network depths.

(a) Starting Channel 8 results			(b) Starting Channel 16 results		
Layers	L1	MSE	Layers	L1	MSE
3	64.52	72.58	3	74.09	72.76
4	63.06	59.64	4	65.73	64.34

Table 4.4: Linear assignment matching accuracy using **DoubleConv ConvBlocks** without GNN. Results show 512-dimensional MLP features obtained from multi-layer perceptron with L1 and MSE loss across different starting channel sizes and network depths.

(a) Starting Channel 8 results			(b) Starting Channel 16 results		
Layers	L1	MSE	Layers	L1	MSE
3	8.53	11.13	3	14.53	11.38
4	5.38	6.02	4	4.30	7.69

Table 4.5: Linear assignment matching accuracy using **DoubleConv ConvBlocks** with GNN integration. Features combine 512 MLP features with 8 geometric and structural features (total 520) processed by Graph Attention Network across different starting channel sizes and network depths.

(a) Starting Channel 8 results			(b) Starting Channel 16 results		
Layers	L1	MSE	Layers	L1	MSE
3	57.11	62.17	3	67.81	63.98
4	54.64	44.71	4	49.75	51.16

Graph Matching reached 92.00%, and the RAPCAT Method achieved even higher at 95.40%.

Although our method still has room for improvement in terms of accuracy, our work establishes a strong baseline for what can be achieved with an end-to-end deep learning framework. Unlike traditional methods that rely heavily on hand-crafted features, our approach offers several key advantages:

- **End-to-end learning:** The entire feature extraction and matching process can be jointly optimized
- **Graph structure awareness:** Through GNN integration, it can fully utilize spatial relationships between cells

Table 4.6: The best result comparing with previous methods.

	Acc.
Active Graph Matching Kainmueller et al. [2013]	92.00
RAPCAT Method Li et al. [2024]	95.40
Ours	74.09

- **Scalability:** The framework design allows flexible adjustment of network architecture and feature dimensions

Future research directions include further optimization of network architectures, exploration of more advanced Graph Neural Network variants, and integration of multi-modal information to enhance matching accuracy.

5 Conclusion and Outlook

5.1 Conclusion

This research introduces a novel, end-to-end deep learning framework for 3D microscopy cell matching in *C. elegans*, integrating 3D Convolutional Neural Networks (CNNs) with Graph Attention Networks (GATs). Our model achieves a matching accuracy of 74.09%, demonstrating the viability of a pure deep learning approach for this complex biological correspondence problem. While a performance gap to state-of-the-art methods remains, our work establishes a robust baseline and provides critical insights into model architecture and feature representation for automated cellular annotation.

5.1.1 Key Contributions

Our primary contributions are threefold:

1. **An Integrated End-to-End Architecture:** We successfully designed and validated an end-to-end pipeline that jointly optimizes feature extraction and cell matching. By combining a 3D CNN for learning local cellular morphology from voxel data with a GAT for modeling global spatial context, our framework eliminates the need for manual feature engineering and captures complex inter-cellular relationships. The remarkable 56.03% accuracy improvement after integrating the GNN underscores the critical importance of contextual information in this task.
2. **Multi-modal Feature Fusion:** Our approach effectively fuses high-dimensional, learned deep features (512-D) with low-dimensional, explicit geometric features (8-D). This hybrid strategy leverages the complementary strengths of both data-driven and knowledge-based representations, creating a more discriminative and biologically meaningful feature space for accurate cell correspondence.
3. **Systematic Design Principles for Cell Matching Models:** Through extensive ablation studies, we provide valuable architectural and optimization guidelines. We empirically demonstrate the superiority of Residual connections over simple convolutional blocks and show that a 3-layer network strikes a better balance between representational power and detail preservation than a deeper 4-layer variant. Furthermore, our analysis confirms that L1 loss consistently outperforms MSE loss for this regression-style matching task, offering a clear recommendation for future work in this domain.

5.1.2 Limitations

Despite its contributions, this study has several limitations:

- **Performance Gap:** Our 74.09% accuracy, while significant, trails state-of-the-art methods like RAPCAT (95.40%) and Active Graph Matching (92.00%). This highlights the remaining challenges in developing deep learning models that can fully compete with specialized, non-end-to-end pipelines.
- **Computational Constraints:** The experiments were conducted on a limited subset of the *C. elegans* dataset due to significant computational resource requirements. This restriction may affect the model’s generalization performance on the full 558-cell atlas and limits the scale of our evaluation.
- **Incomplete Exploration of Rotation Invariance:** The planned implementation of rotation-invariant features using group-equivariant convolutions was not fully realized due to technical complexity and resource limitations. Consequently, the potential benefits of rotation invariance for cell matching remain an open question.

5.2 Outlook and Future Work

Our exploration into rotation-invariant deep learning, particularly through the partial implementation of CubeNet, has illuminated both the immense potential and the significant technical hurdles of applying principled geometric deep learning to 3D biological image analysis. The primary challenges—prohibitive computational costs and difficulties in integrating with modern network architectures—necessitate a clear roadmap for future research. This section outlines this roadmap, moving from immediate priorities to long-term strategic directions.

5.2.1 Addressing Current Limitations and Short-Term Priorities

The immediate priority is to establish a complete and robust baseline for group equivariant methods. Our initial CubeNet implementation revealed two critical bottlenecks:

- **Computational and Memory Overheads:** The core limitation is the substantial computational overhead of group convolutions. Employing the T4 rotational group (12 elements) increased memory requirements by a factor of 12, making batch sizes larger than 32-64 impractical even on an NVIDIA A100 GPU. This memory bottleneck, caused by the additional group dimension in feature maps, directly prevented the loading and processing of all 558 cell features simultaneously, thereby hindering downstream matching tasks like the Hungarian algorithm.

- **Feature Compression Incompatibility:** A fundamental architectural challenge emerged when integrating group equivariant features with standard compression techniques like NdLinear. Conventional pooling and linear layers disrupt the group structure, nullifying the theoretical benefits of equivariance.

Therefore, the most critical short-term goal is to refine the CubeNet implementation, possibly by investigating sparse or adaptive group convolution techniques, to achieve a functional baseline. This will enable a meaningful performance comparison against traditional 3D CNNs and provide crucial insights for subsequent steps.

5.2.2 Long-Term Research Directions

Building on our findings, we propose three primary avenues for long-term research to significantly advance the state-of-the-art in automated cell matching.

Exploring Alternative Architectures for Rotation Invariance

While refining CubeNet is a priority, exploring more advanced and potentially more efficient frameworks is crucial.

- **Euclidean Neural Networks (e3nn):** As a leading alternative, e3nn offers a compelling approach grounded in geometric algebra and representation theory Geiger and Smidt [2022]. It provides stronger theoretical guarantees for 3D rotation and reflection equivariance and may be more computationally efficient than explicit group convolution enumeration Thomas et al. [2018], Fuchs et al. [2020]. However, its implementation is non-trivial, demanding expertise in geometric deep learning and careful management of memory-intensive spherical harmonic and tensor field operations Weiler et al. [2018]. A key long-term objective is to develop an e3nn-based model and perform a direct comparative analysis against the CubeNet baseline.
- **Other Invariant Frameworks:** Further investigation into steerable CNNs and capsule networks could offer alternative pathways to achieving robust rotation invariance, potentially striking a different balance between theoretical guarantees and computational cost.

Self-Supervised with Contrastive Learning

To address the limited availability of annotated biological data, self-supervised methods like Contrastive Learning (CL) present a powerful paradigm. Building on frameworks like SimCLR Chen et al. [2020], future work can proceed along four axes:

- **Domain-Specific Data Augmentations:** A primary direction is to design principled augmentations for 3D microscopy data. This includes realistic elastic deformations, 3D rotations, and intensity variations to model imaging

artifacts and biological variability, moving beyond the natural-image augmentations (e.g., color distortion) used in the original SimCLR.

- **Optimized Architectures for Cell Features:** The non-linear projection head in CL can be adapted to learn features optimized for our downstream tasks. For instance, the head could be designed to explicitly disentangle morphological features (for the 3D CNN) from positional information (for the GNN), leading to more specialized feature extractors.
- **Efficient Training on 3D Data:** Standard CL requires large batch sizes, which is prohibitive for 3D data. A vital research direction is to develop or adapt CL frameworks that are efficient with smaller batches, perhaps using memory banks or sophisticated negative sampling, to make pre-training on large-scale unlabeled *C. elegans* datasets feasible.
- **Graph-based Contrastive Learning:** A truly innovative frontier is to extend CL to incorporate structural context. Instead of contrasting individual cell volumes, a graph-based CL framework could contrast entire cellular neighborhoods (sub-graphs). This would train the feature extractor to be inherently aware of the local spatial context vital for the GNN, creating a powerful synergy within the pipeline.

Advanced Computational and Resource Management Strategies

Overcoming the computational bottlenecks identified in this work is essential for any future progress. Future implementations must strategically employ advanced optimization techniques, such as:

- **Gradient Checkpointing:** To trade computation for memory, significantly reducing memory requirements during backpropagation [Chen et al. \[2016\]](#).
- **Mixed-Precision Training:** Using FP16 to reduce memory footprint and accelerate training on compatible hardware [Micikevicius et al. \[2017\]](#).
- **Distributed Computing:** Leveraging frameworks to parallelize operations across multiple GPUs, enabling the processing of larger models and datasets [Li et al. \[2014\]](#).

Part I

Appendix

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Erklärung:

Ich versichere, dass ich diese Arbeit selbstständig verfasst habe und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Heidelberg, den July 24, 2025

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Declaration:

I hereby confirm that I wrote this work independently and did not use any sources other than those indicated.

Heidelberg, July 24, 2025

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